

First steps in IoT with LoRaWAN

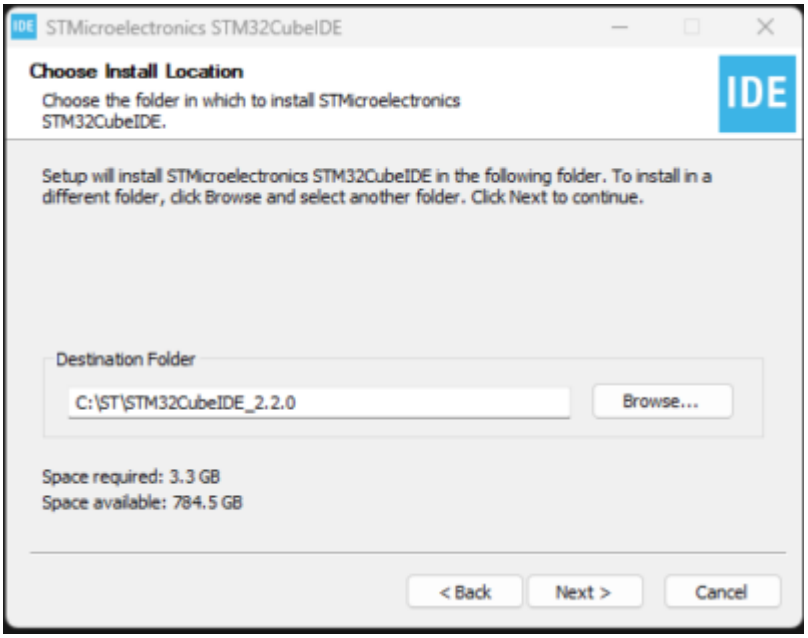
Development environment

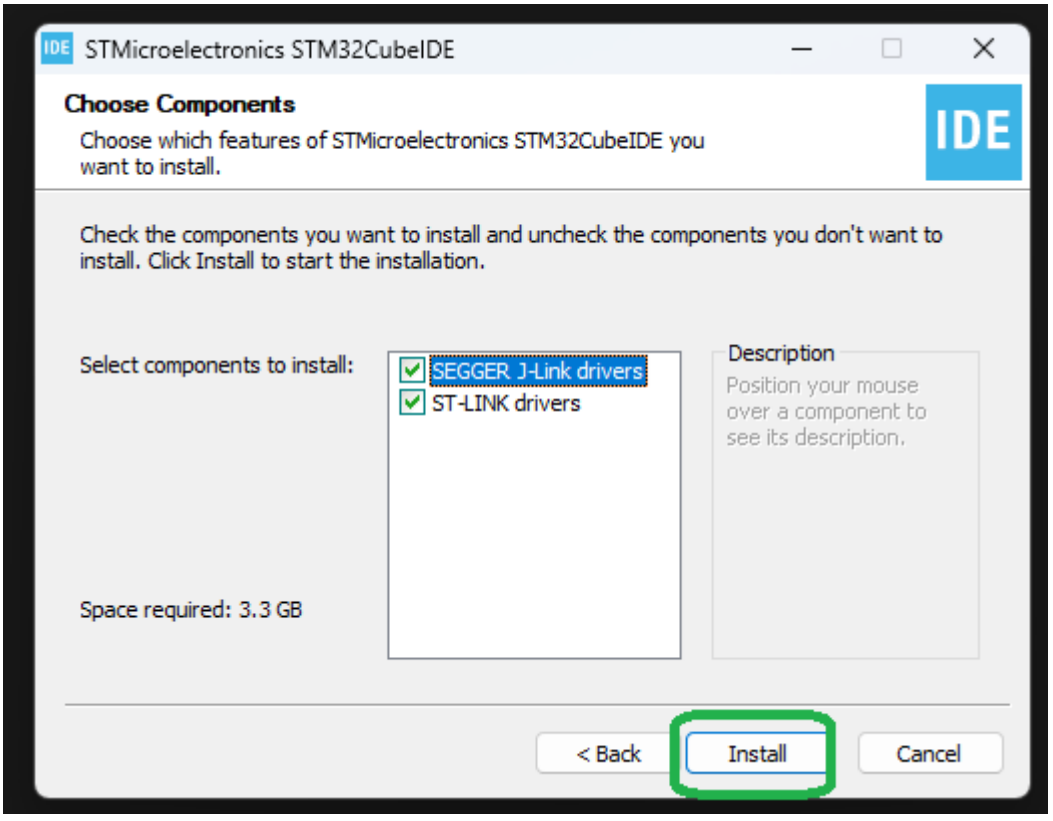
- Hardware/Firmware
  - Gateway(optional): NUCLEO-F746ZG + LRWAN GS HF1 LoRa
  - Development board:STM32WL55JC1 : STM32WL55JCIx Firmware: V1.3.1 + driver
  - Apiar IoT PCB: Available soon.
- *Software:*
  - STMCubeIDE(V:2.2.0)
  - STM32CubeMX(V:6.18.1)
  - STMCubeWL(V1.6.0/1): Library for STM32WL55JC
  - STM32WL55JCIx development board Firmware: V1.3.

Preparing the environment (toolchain)

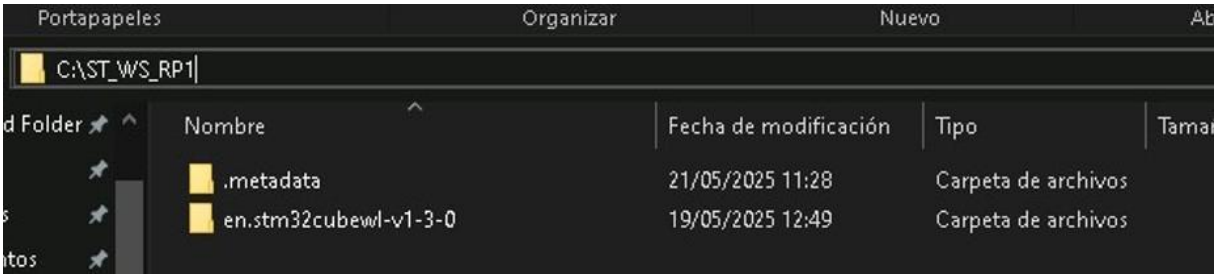
It is recommended to keep the same folder path and follow each step exactly.

- Download and install the integrated development environment (IDE) ”STM32CubeIDE”.

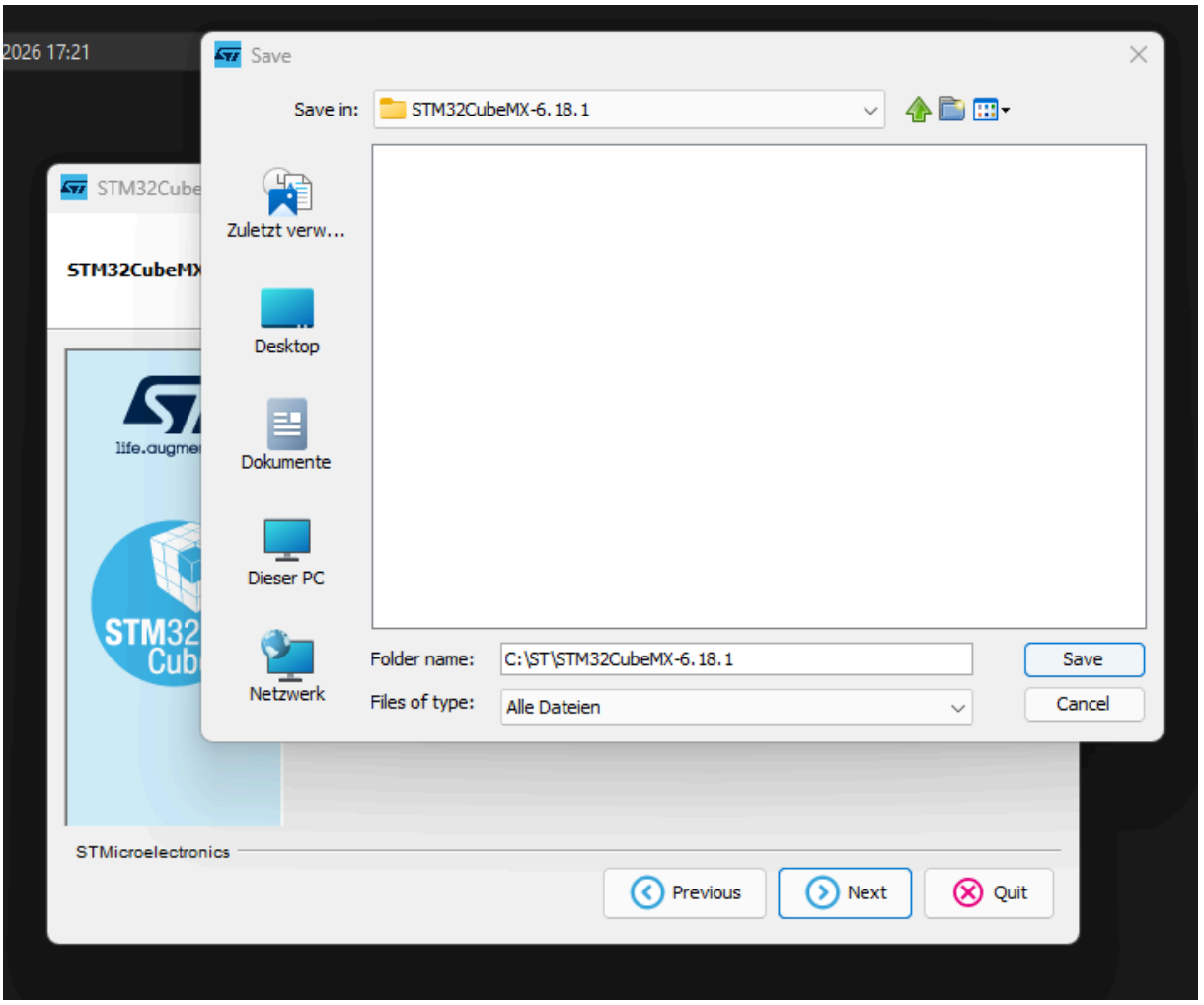


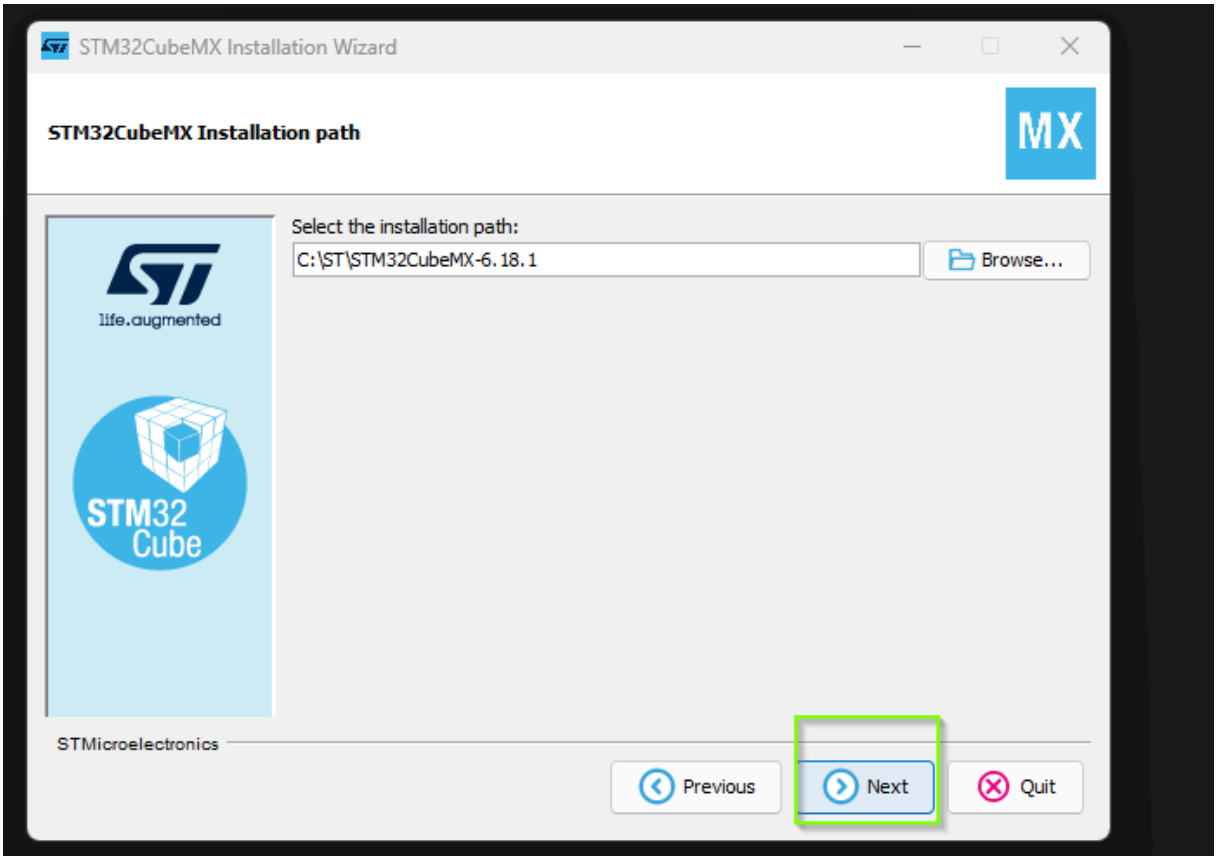


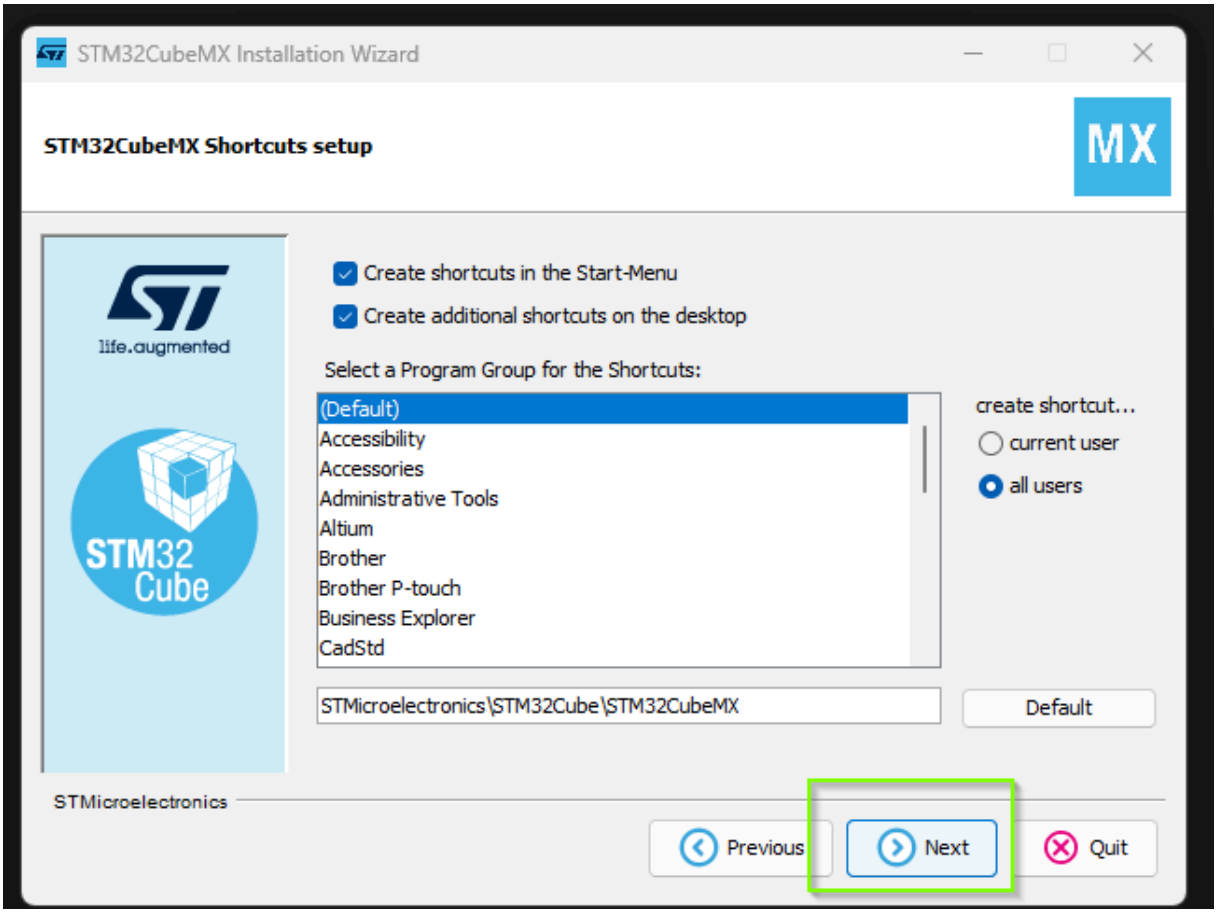
- Create the folder "C:\ST_WS_RP1".
  It is recommended to use the same folder and the same path.



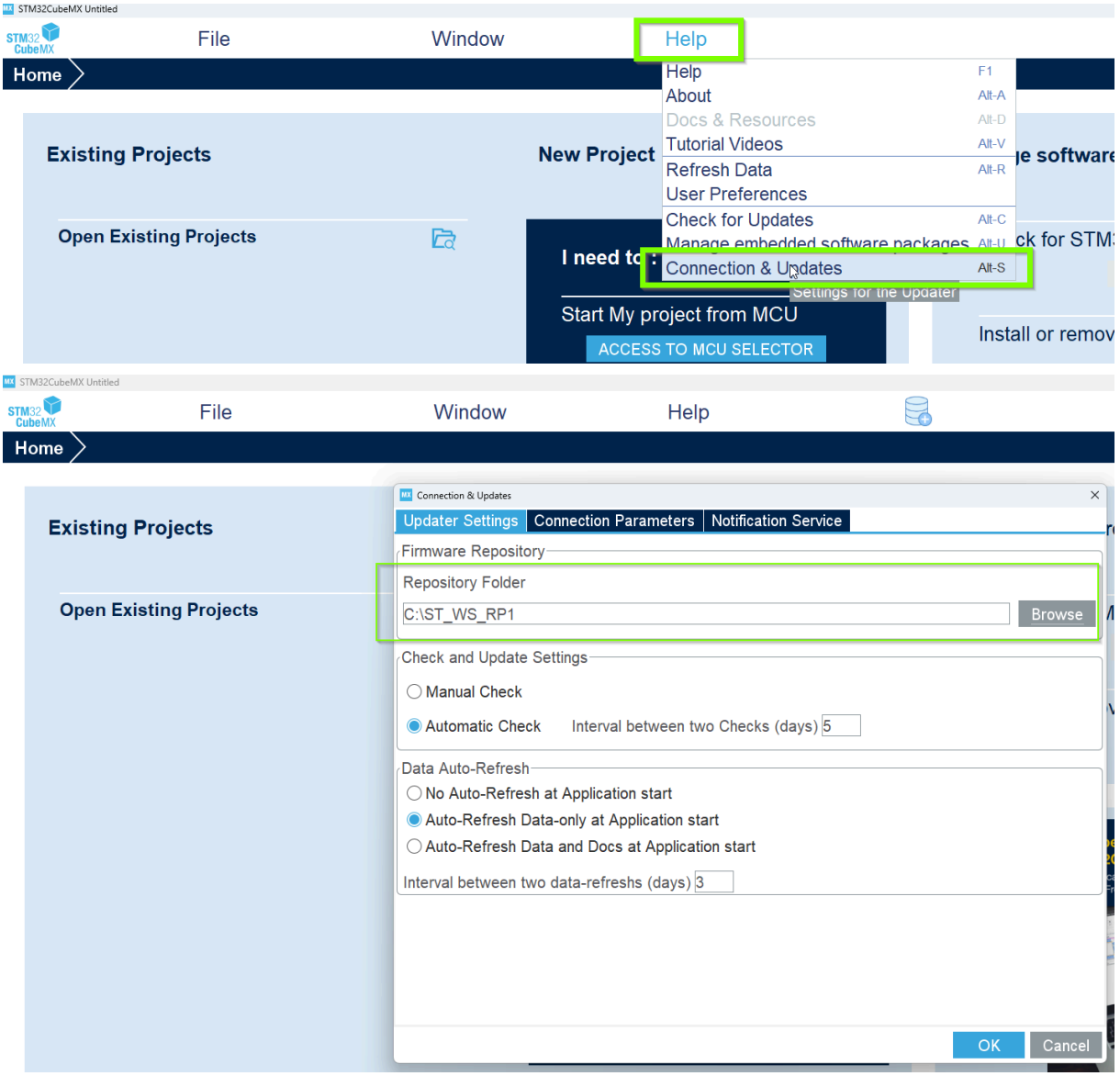
- Create the folder "C:ST\STM32CubeMX-6.18.1",Download and install STM32CubeMX(V:6.18.1)

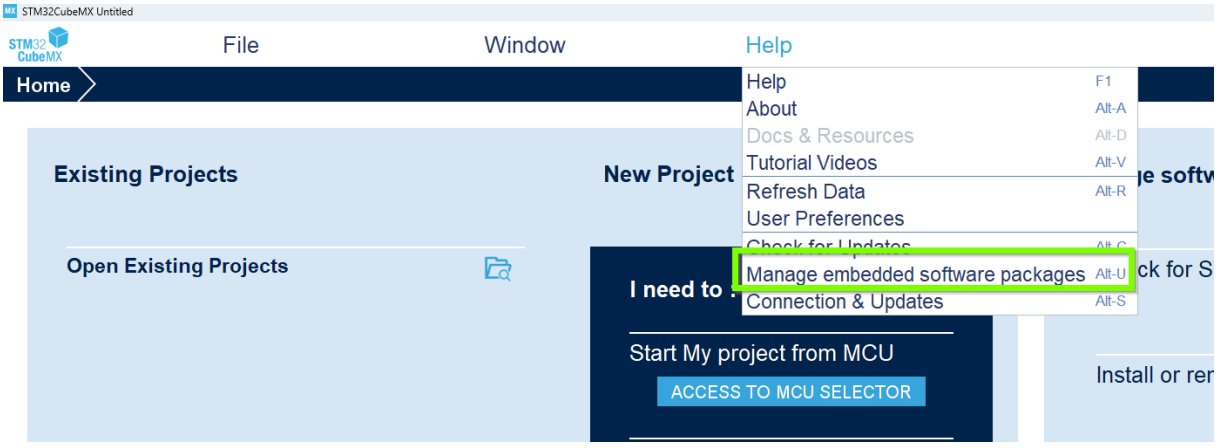


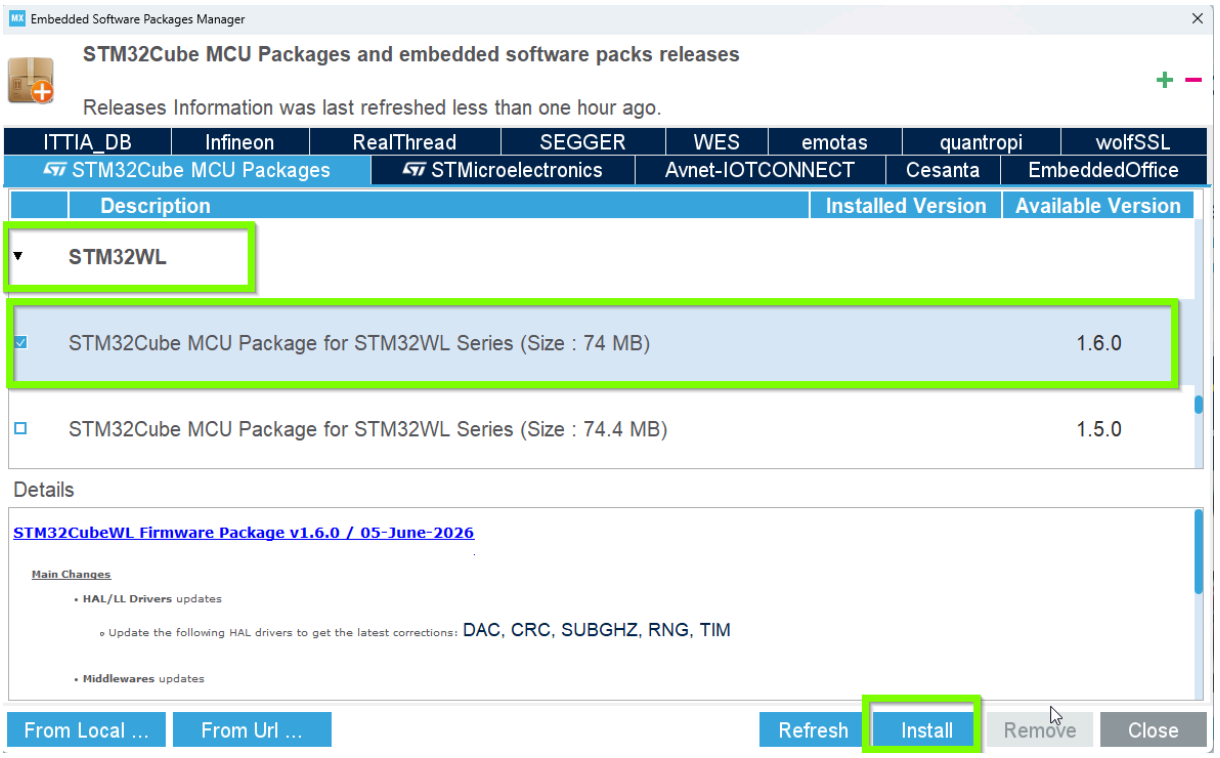

STM32CubeMX Installation Wizard
STM32CubeMX Installation path
MX
life.augmented
STM32 Cube
Select the installation path:
C:\ST\STM32CubeMX-6.18.1
Browse...
STMicroelectronics
Previous
Next
Quit


STM32CubeMX Installation Wizard
STM32CubeMX Shortcuts setup
MX
life.augmented
STM32 Cube
Create shortcuts in the Start-Menu
Create additional shortcuts on the desktop
Select a Program Group for the Shortcuts:
(Default)
Accessibility
Accessories
Administrative Tools
Altium
Brother
Brother P-touch
Business Explorer
CadStd
create shortcut...
current user
all users
STMicroelectronics\STM32Cube\STM32CubeMX
Default
STMicroelectronics
Previous
Next
Quit

- Start STM32CubeMX



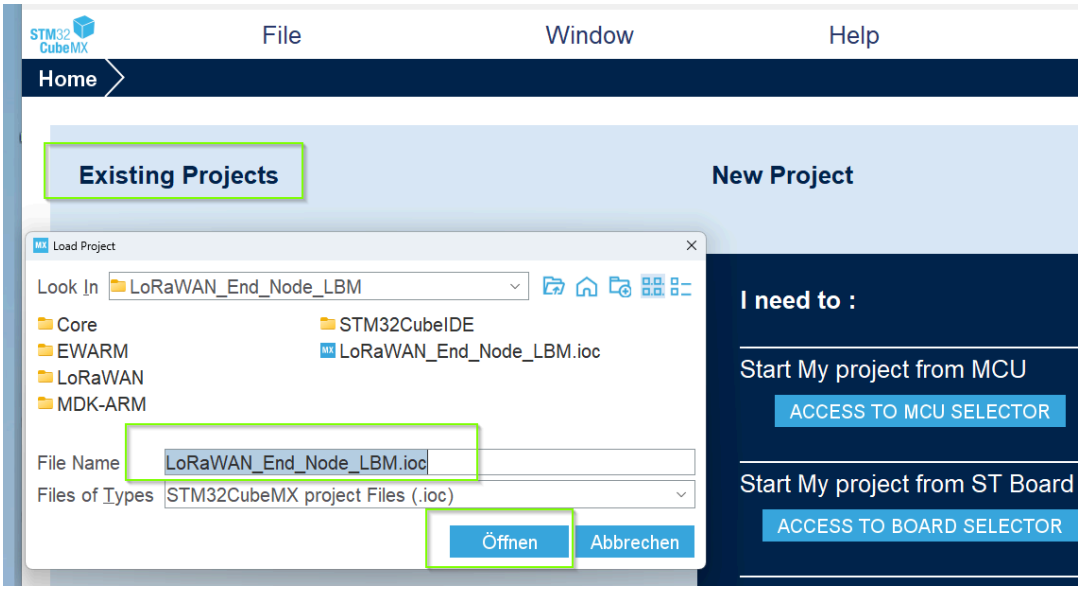

STM32CubeMX Untitled
File
Window
Help
Home
Help F1
About Alt-A
Docs & Resources Alt-D
Tutorial Videos Alt-V
Refresh Data Alt-R
User Preferences
Manage embedded software packages Alt-U
Connection & Updates Alt-S
Existing Projects
New Project
Open Existing Projects
I need to :
Start My project from MCU
ACCESS TO MCU SELECTOR
Install or rer

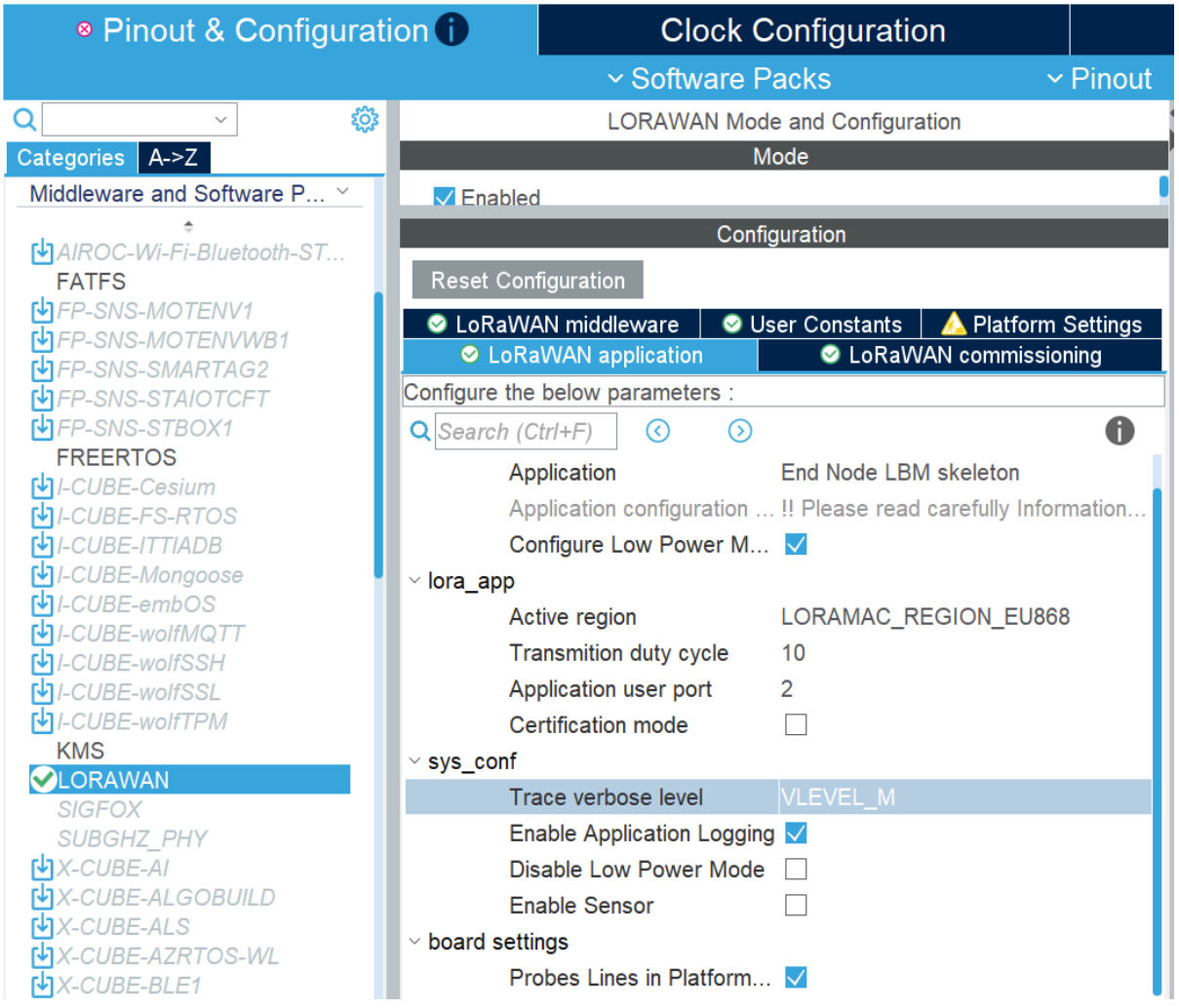

Embedded Software Packages Manager
STM32Cube MCU Packages and embedded software packs releases
Releases Information was last refreshed less than one hour ago.
ITTIA_DB
Infineon
RealThread
SEGGER
WES
emotas
quantropi
wolfSSL
STM32Cube MCU Packages
STMicroelectronics
Avnet-IOTCONNECT
Cesanta
EmbeddedOffice
Description
Installed Version
Available Version
STM32WL
STM32Cube MCU Package for STM32WL Series (Size : 74 MB) 1.6.0
STM32Cube MCU Package for STM32WL Series (Size : 74.4 MB) 1.5.0
Details
STM32CubeWL Firmware Package v1.6.0 / 05-June-2026
Main Changes
HAL/LL Drivers updates
Update the following HAL drivers to get the latest corrections: DAC, CRC, SUBGHZ, RNG, TIM
Middlewares updates
From Local ...
From Url ...
Refresh
Install
Remove
Close

LoRaWAN configuration:

- Starte: LoRaWAN_End_Node_LBM.ioc

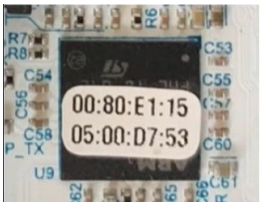
  C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\ProjectsNUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWAN_End_Node_LBM.ioc





LoRaWAN commissioning:

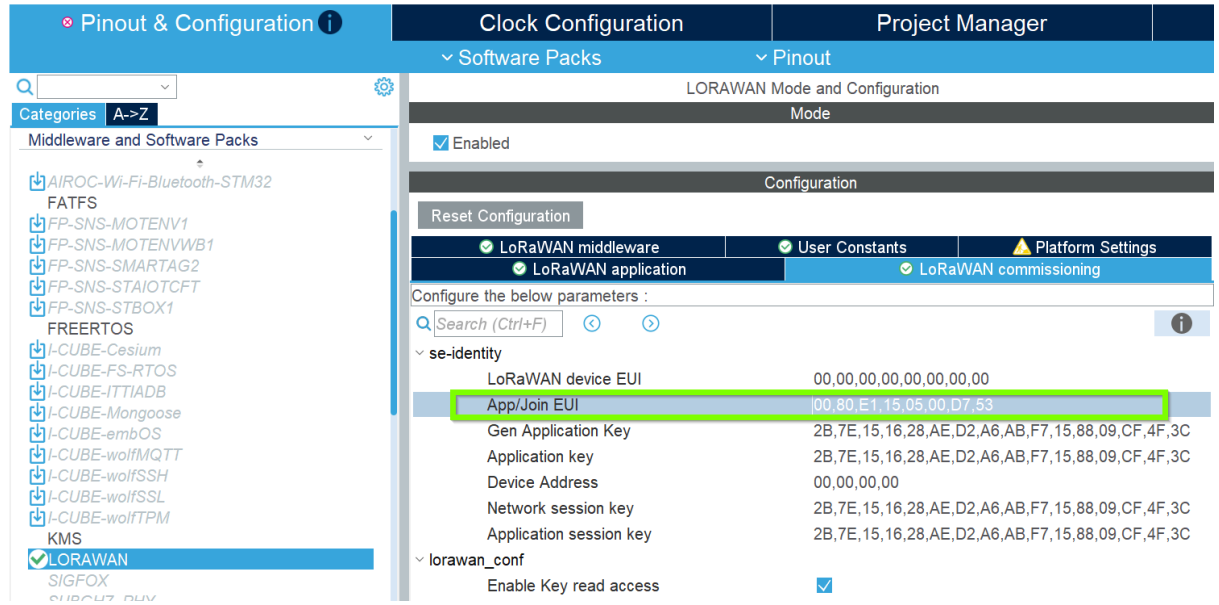
- *“App/Join EUI”*: Can be found on the board.

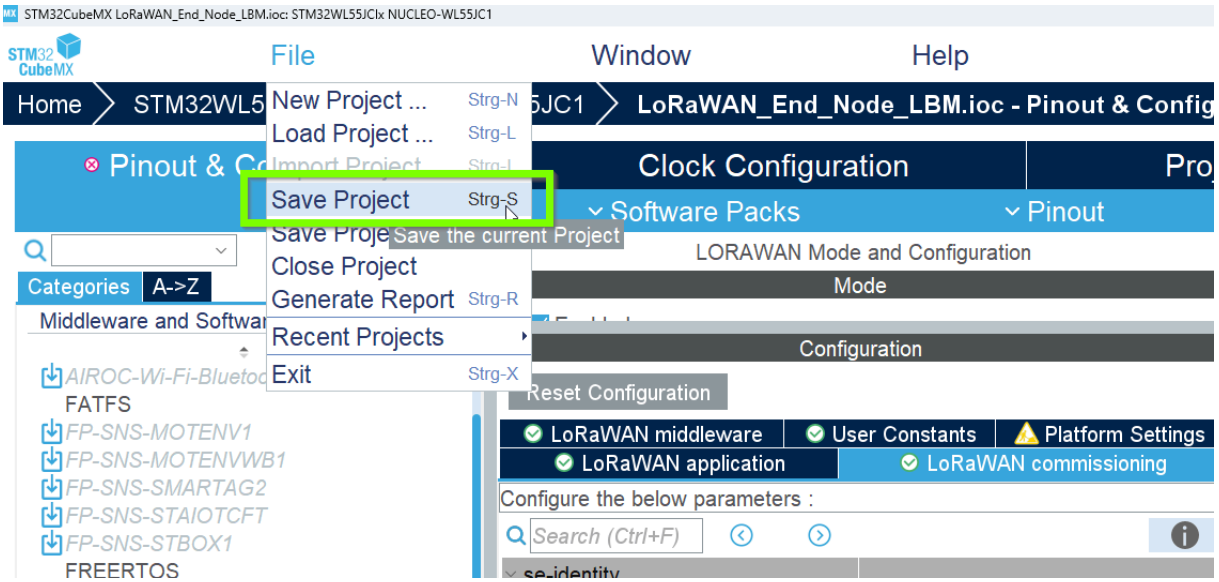


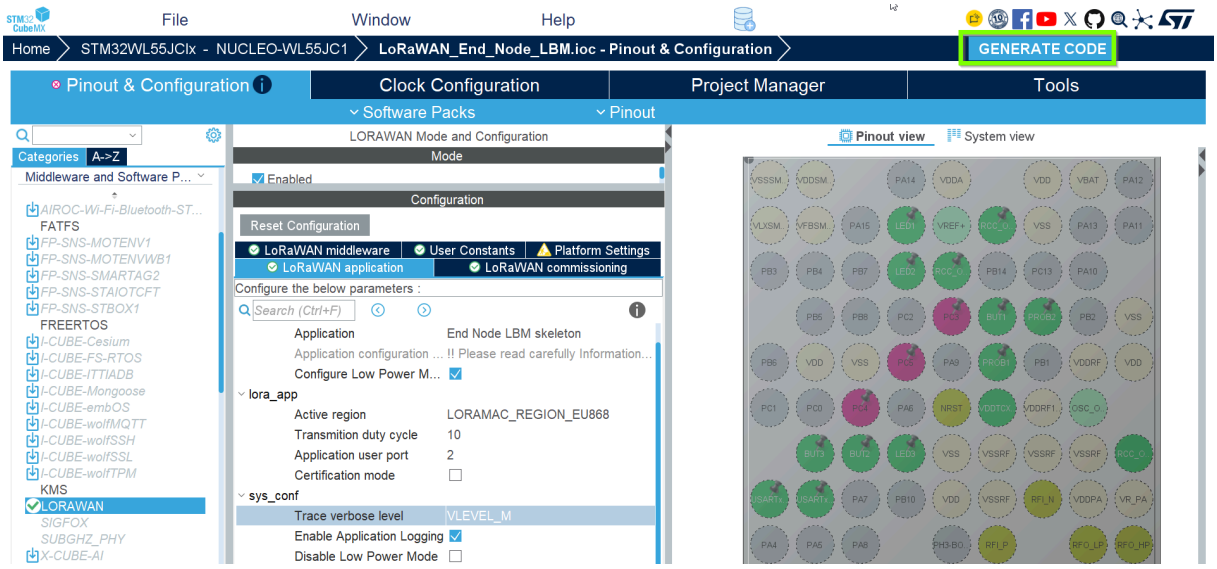
Important: The field "LORAWAN_device EUI" should NOT be defined as "00.00.00.00.00.00.00.00". Choose it yourself.

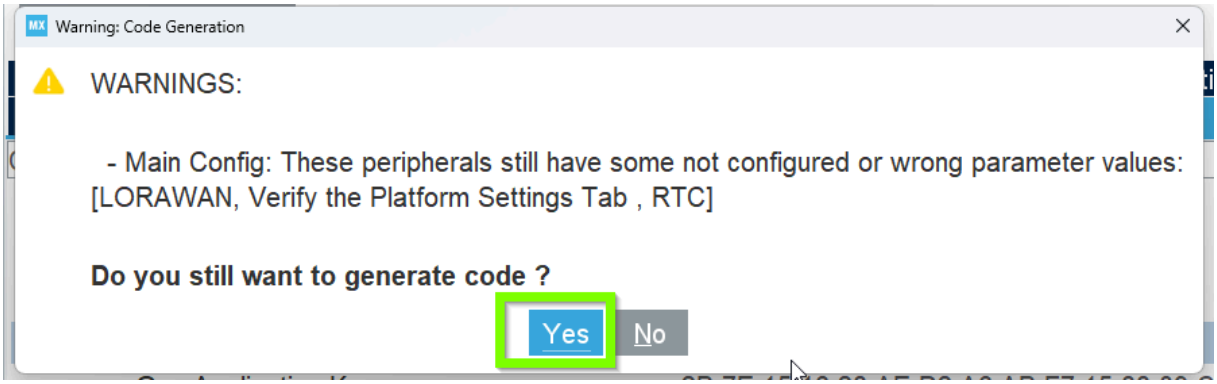
```
/*!
* End-device IEEE EUI (big endian)
* When set to 00,00,00,00,00,00,00,00 DevEui is automatically set
with a value provided by MCU platform
*/
#define LORAWAN_DEVICE_EUI
00,00,00,00,00,00,00,00
```

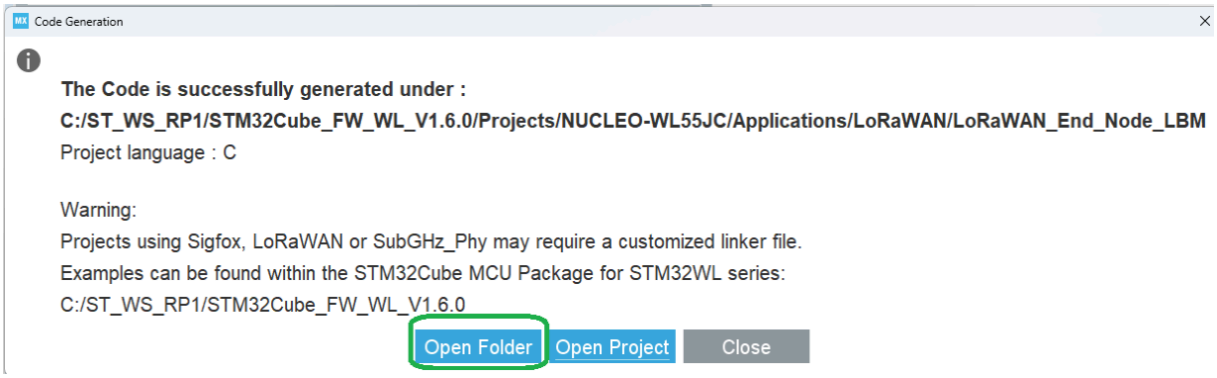
- *“Application key”*: choose yourself



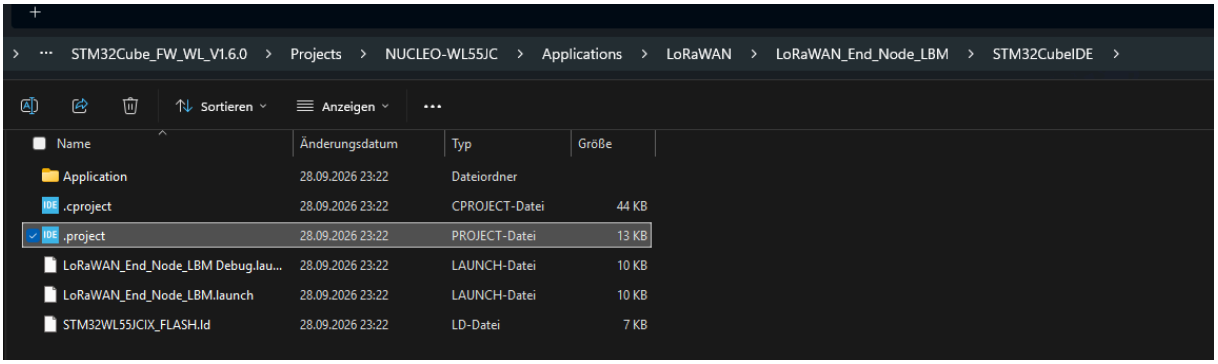




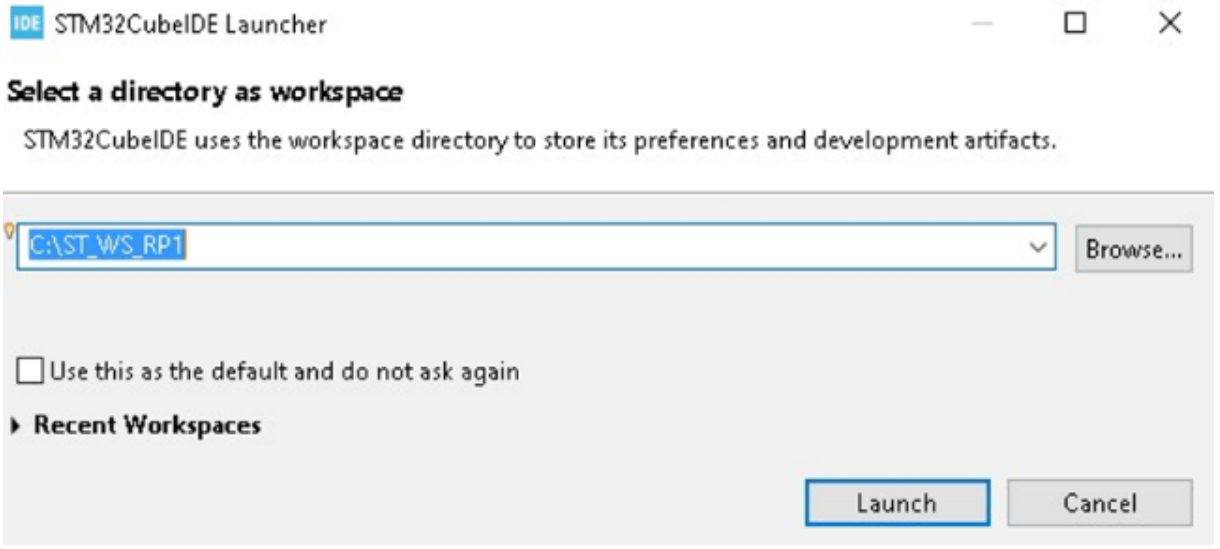


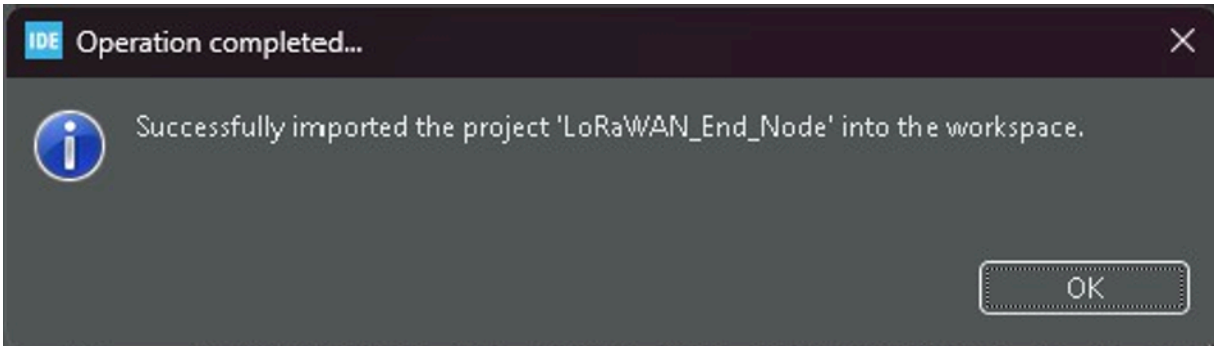


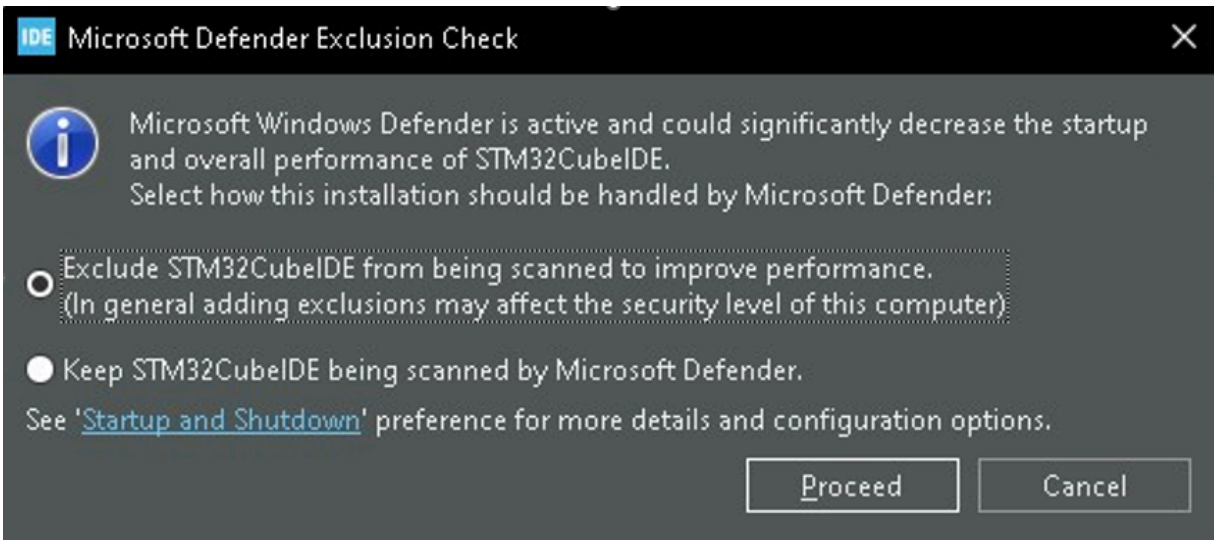
- Start the project with STM32cubeIDE
"C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\STM32CubeIDE\.project"

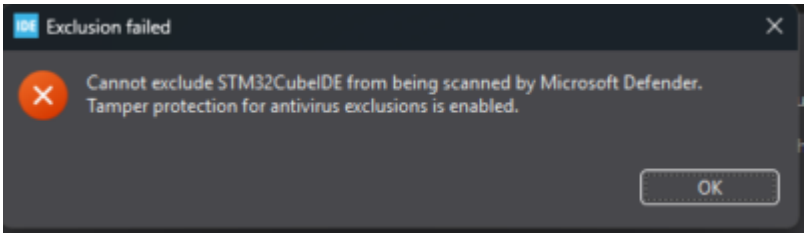


- Select C:\ST_WS_RP1 as the workspace.

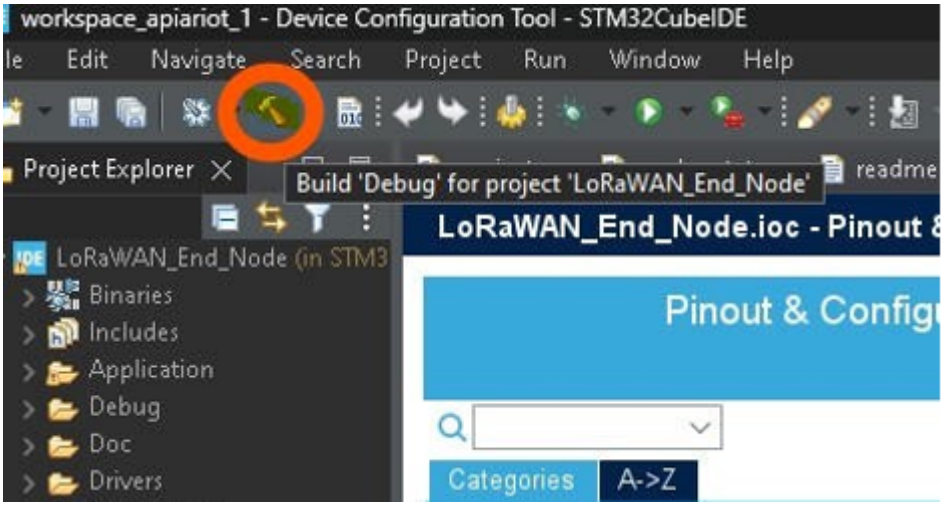


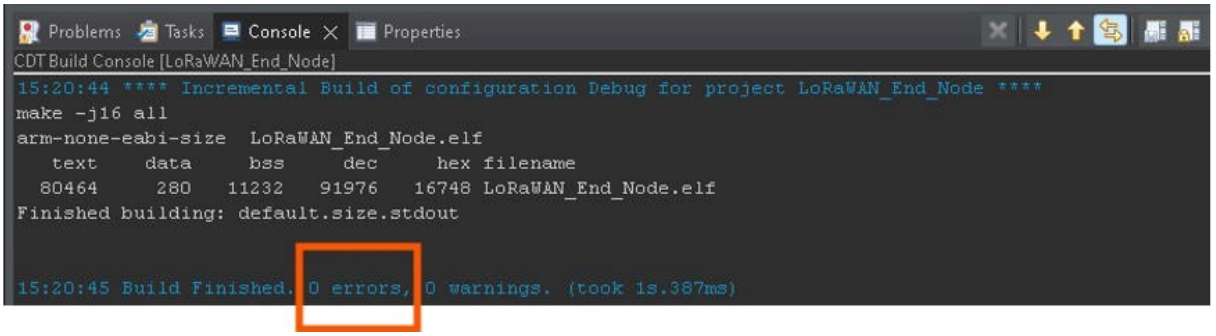






- Compile the project without errors:





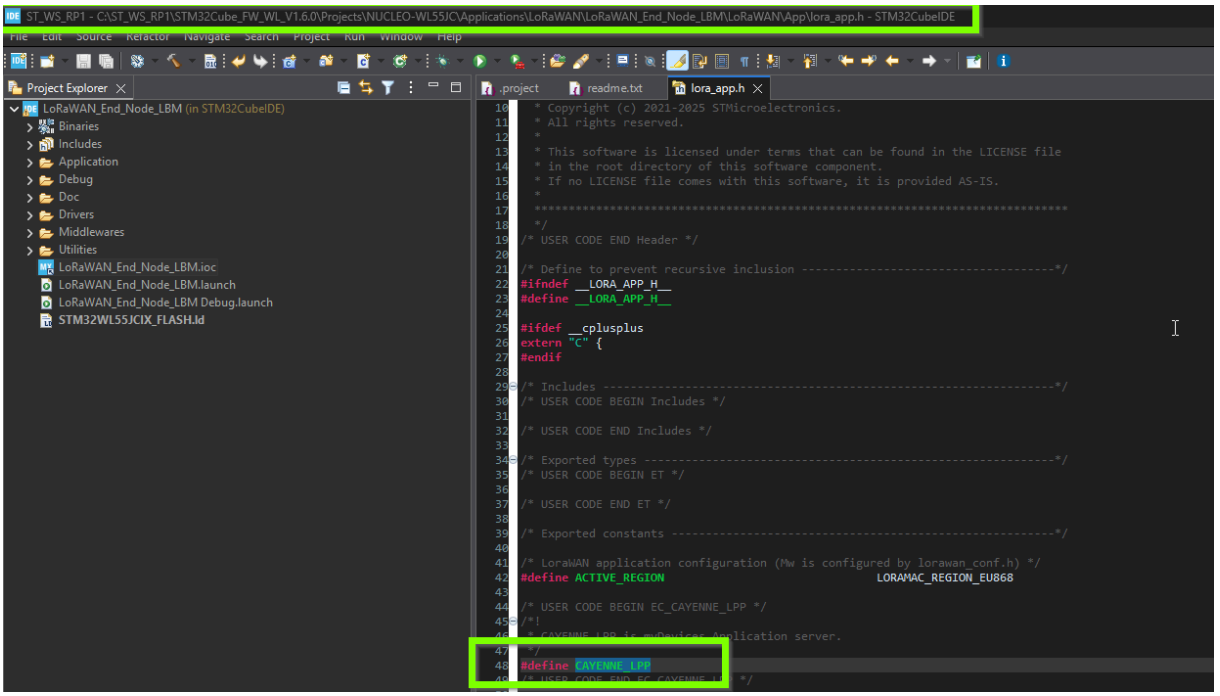
Programming code change

- Activate Cayenne's "Low Power Payload":
    - Uncomment /*#define CAYENNE_LPP in “lora_app.h”*/:
      #define CAYENNE_LPP

"C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWAN\App\lora_app.h""

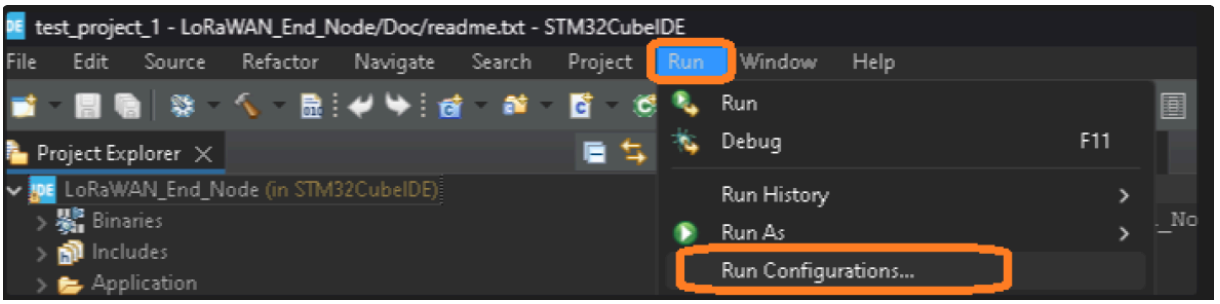
A notice: Cayenne LPP is a compact data format for encoding sensor data for LoRaWAN. More information is available at this link:

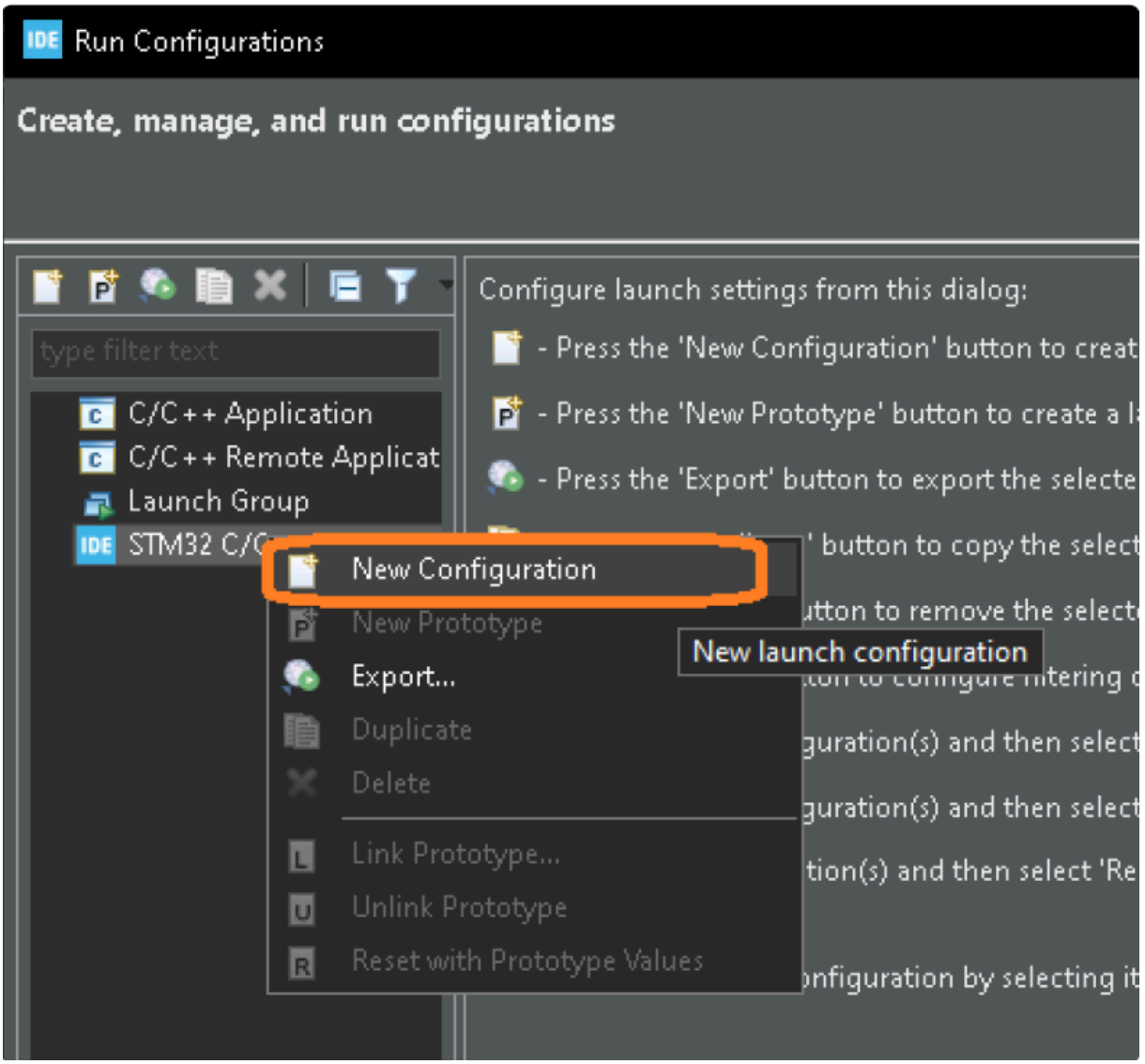
Was ist Cayenne Low Power Payload? 🤔🤷



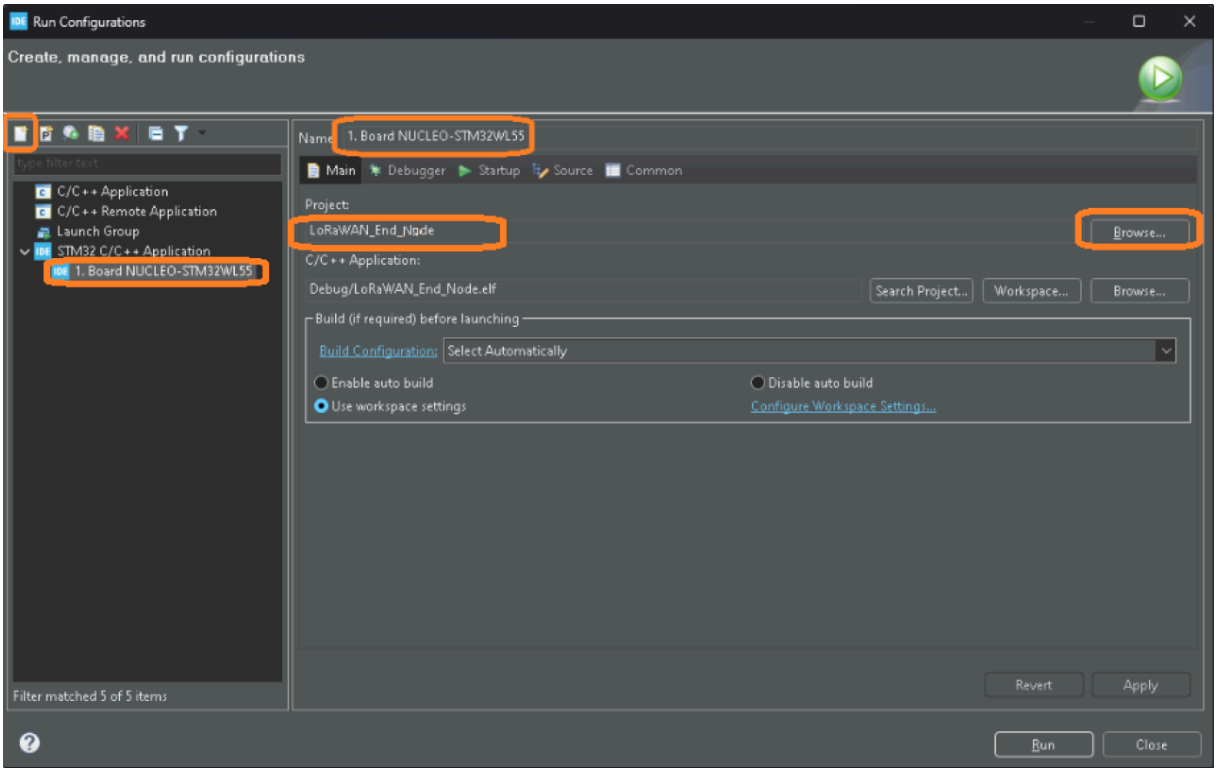
Board settings

- Create a new configuration for the development board.

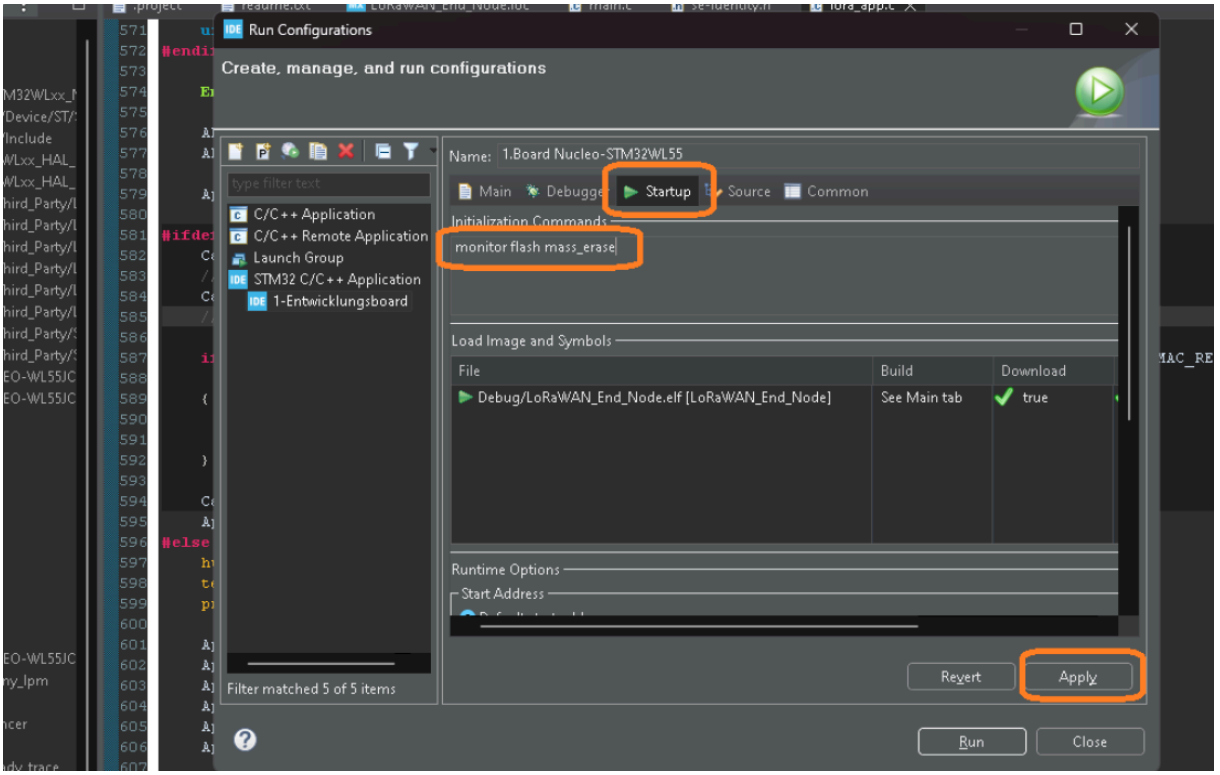




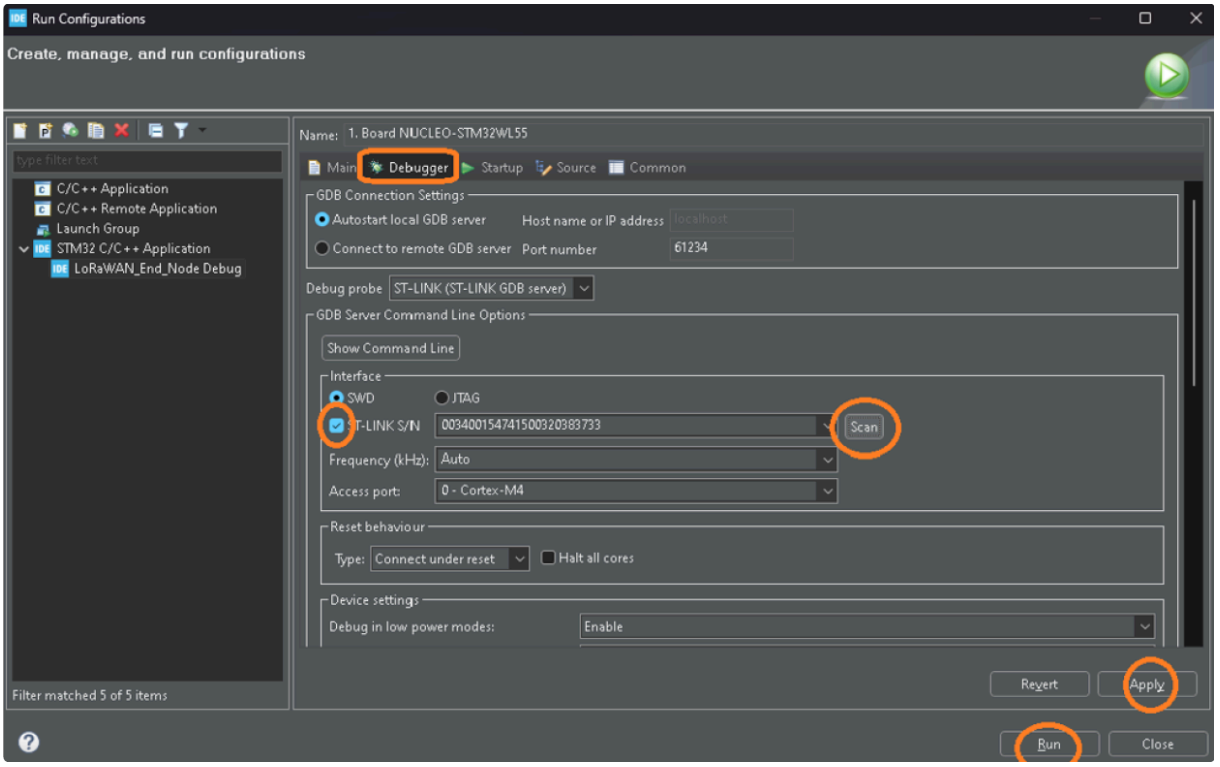
- Name your board NUCLEO-STM32WL55



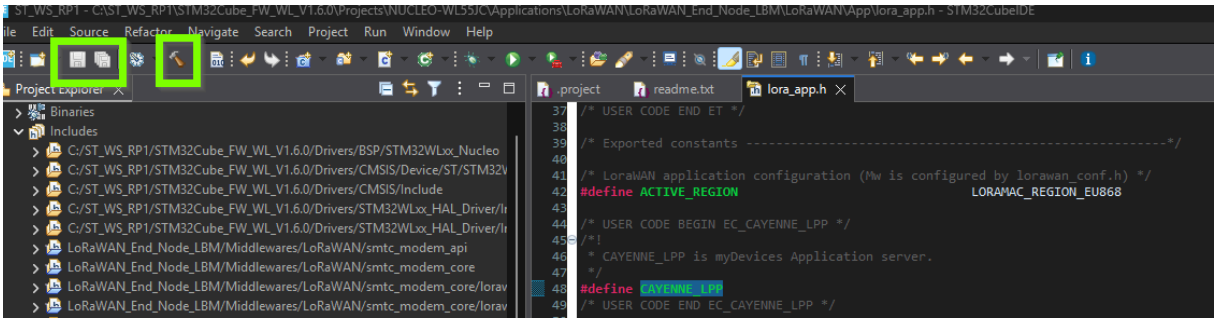
- Add under "*Initialization Commands"* Add the following command:
  "*monitor flash mass_erase"*



- Connect the development board to the PC via USB

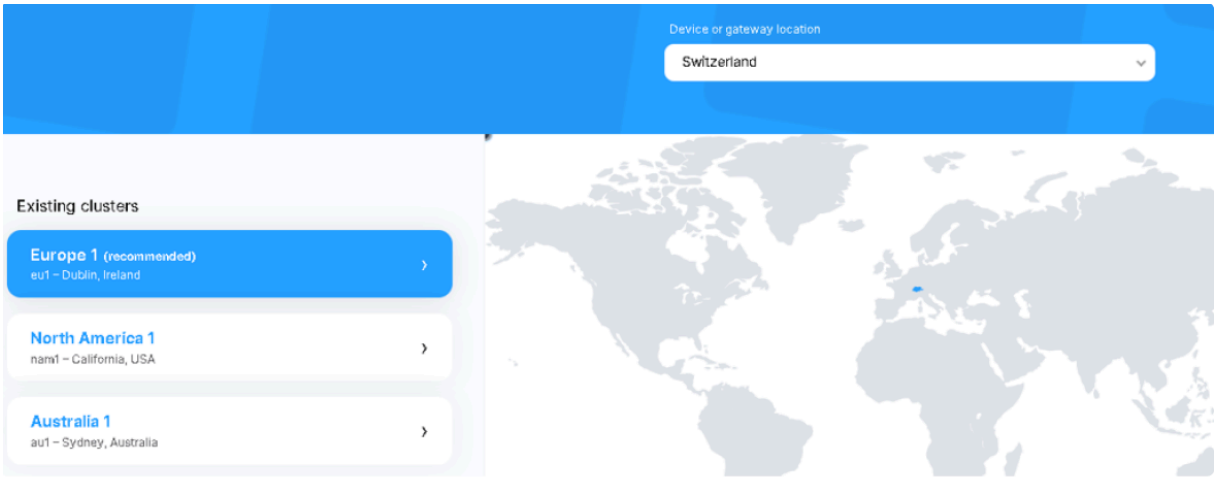


- Save the configuration



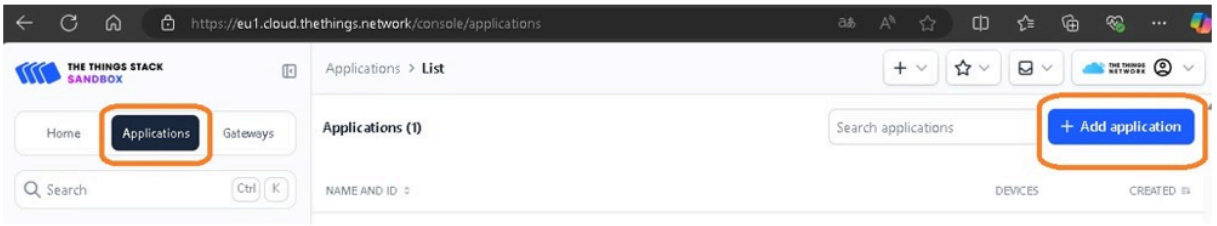
The Thing Network (TTN) / “The Stacks”

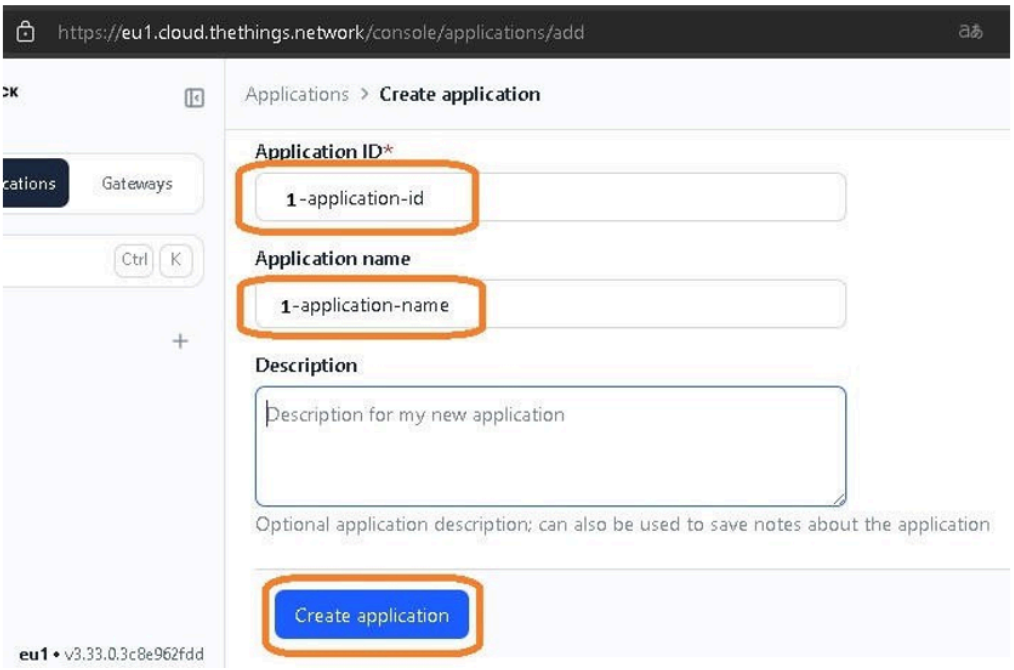
- Create an account to access “The things network”.
- Select Cluster Europe 1 (For Switzerland)

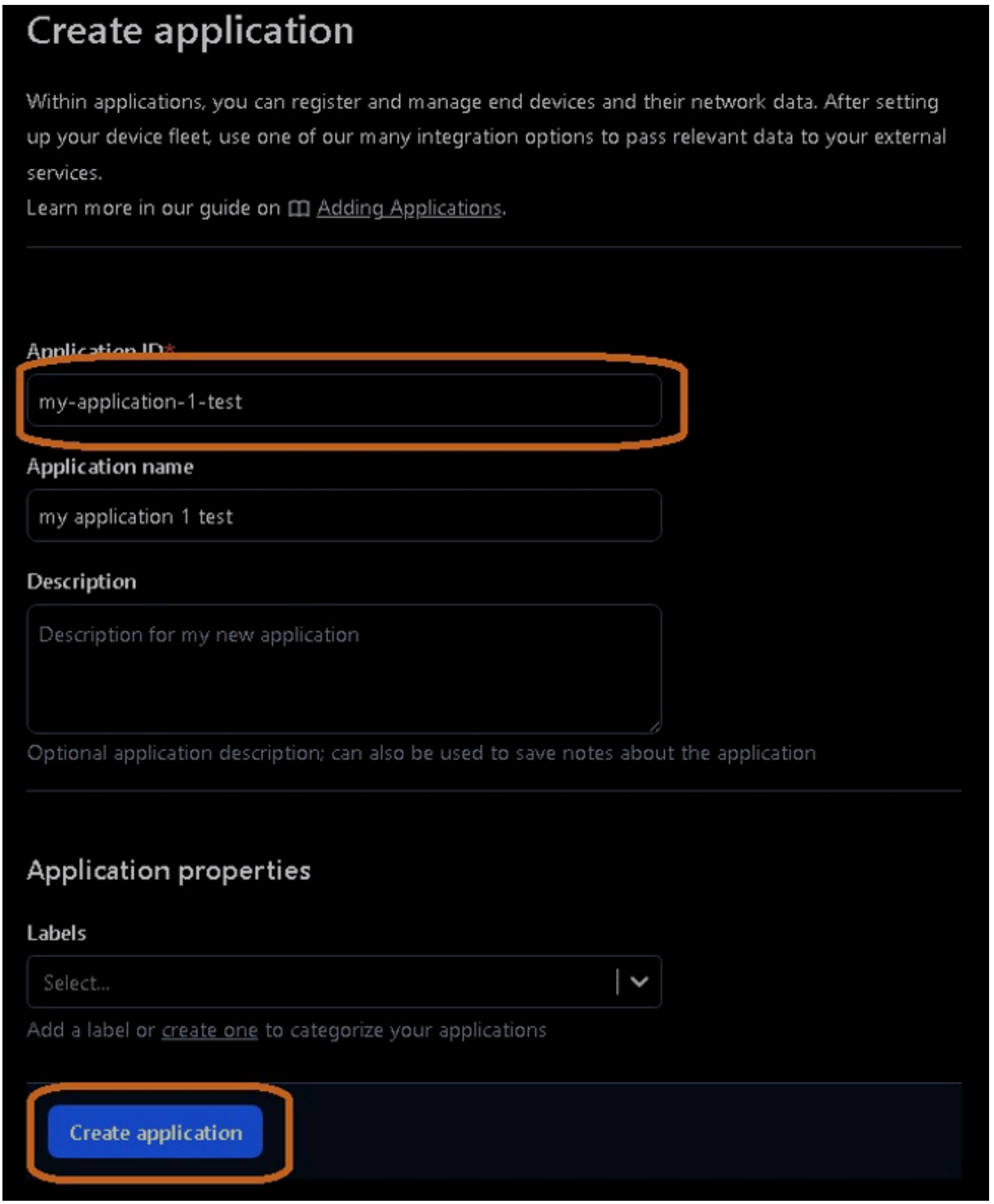


Cluster Für die Schweiz: Europe 1

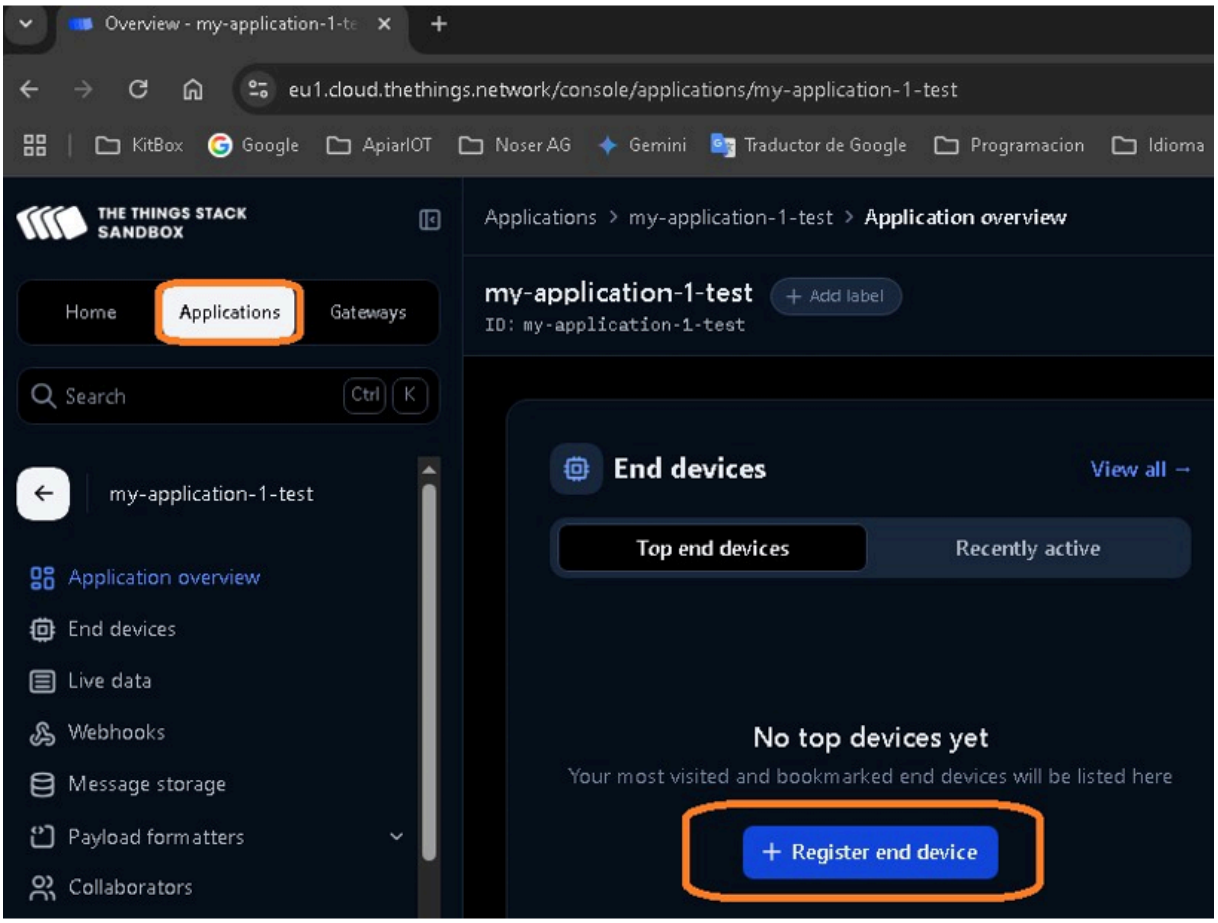
- Create an application



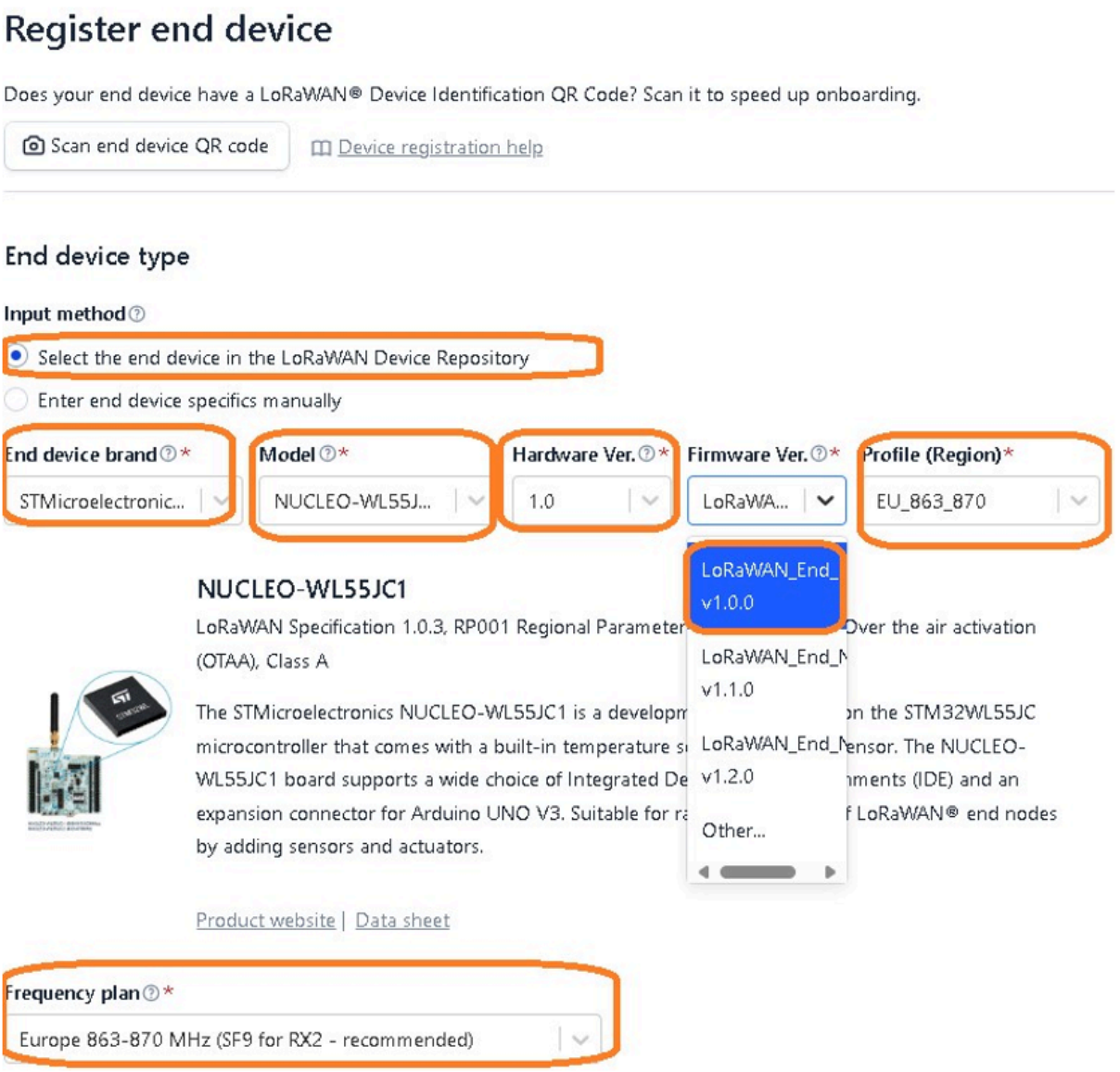



Create application
Within applications, you can register and manage end devices and their network data. After setting up your device fleet, use one of our many integration options to pass relevant data to your external services.
Learn more in our guide on Adding Applications.
Application ID*
my-application-1-test
Application name
my application 1 test
Description
Description for my new application
Optional application description; can also be used to save notes about the application
Application properties
Labels
Select...
Add a label or create one to categorize your applications
Create application

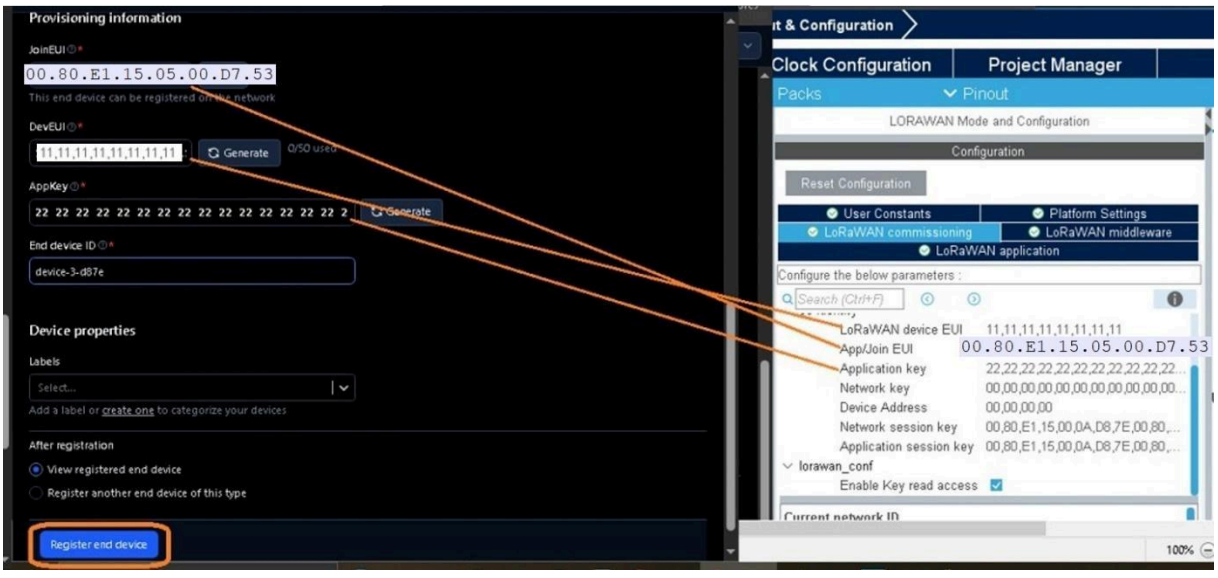
- Register the "end device".

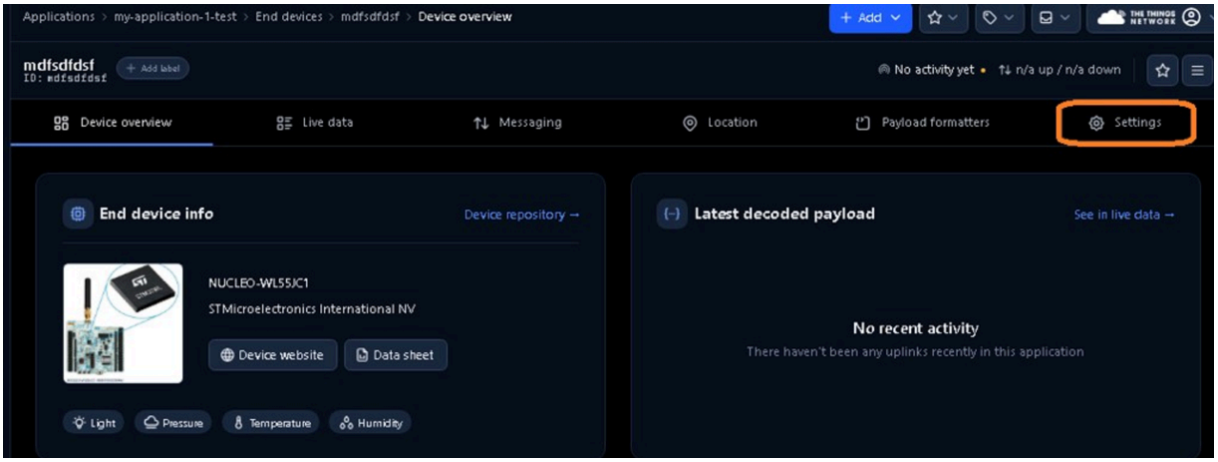


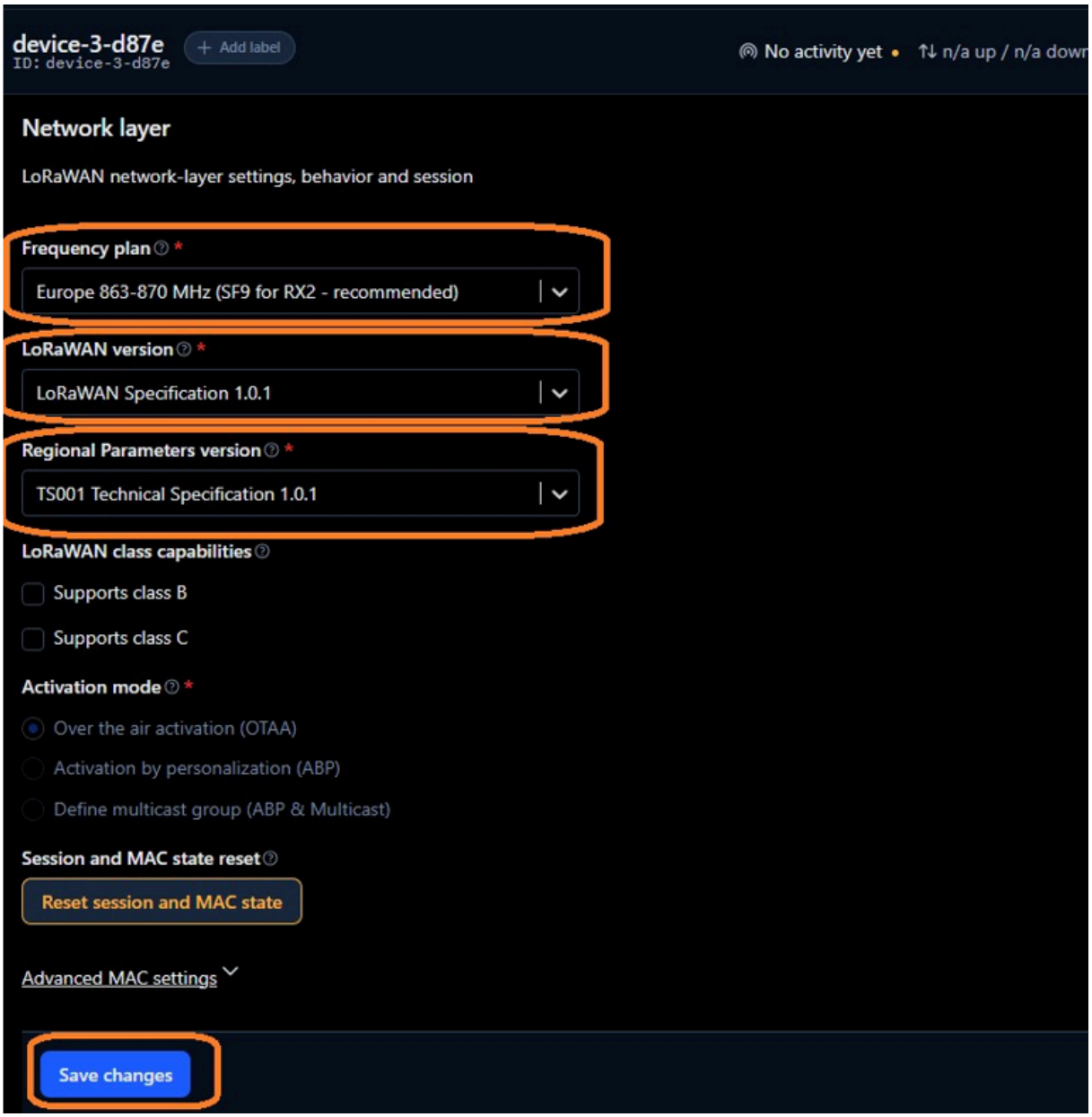
- End device type, End device brand, Model, Hardware version, Firmware version, Profile region, Frequency plan



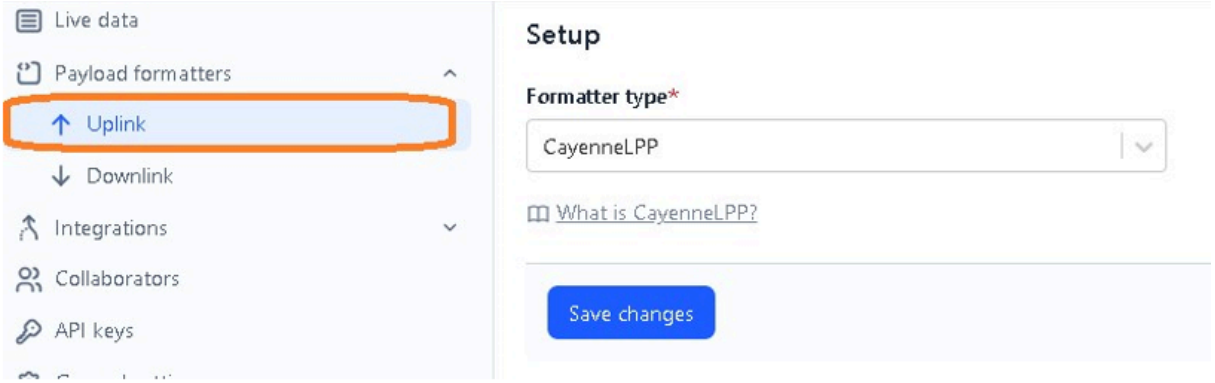
- Search the information on "LoRaWAN commissioning"

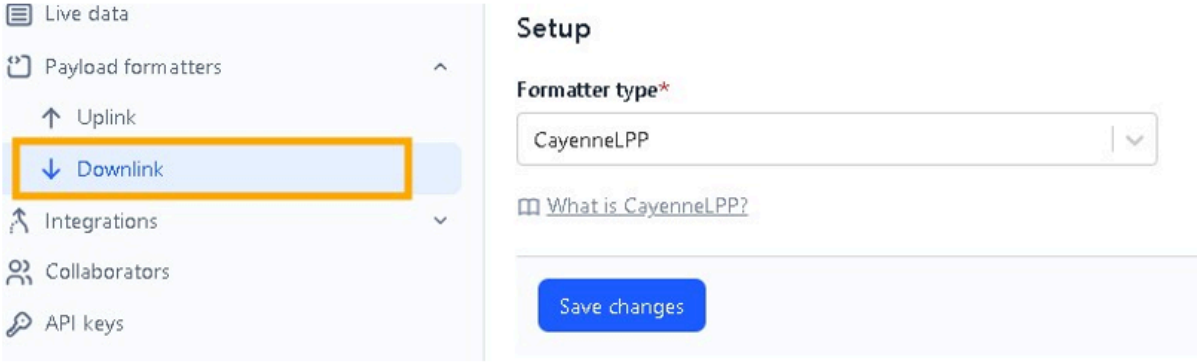



Applications > my-application-1-test > End devices > mdfsdfdsf > Device overview
+ Add
mdfsdfdsf
ID: mdfsdfdsf
+ Add label
No activity yet • ↑↓ n/a up / n/a down
Device overview
Live data
Messaging
Location
Payload formatters
Settings
End device info
Device repository →
NUCLEO-WL55JC1
STMicroelectronics International NV
Device website
Data sheet
Light
Pressure
Temperature
Humidity
Latest decoded payload
See in live data →
No recent activity
There haven't been any uplinks recently in this application

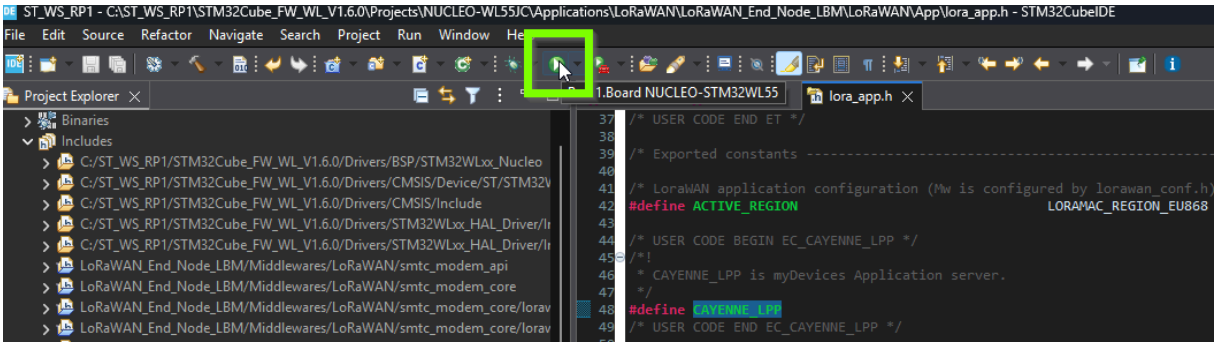

device-3-d87e
ID: device-3-d87e
+ Add label
No activity yet • ↑↓ n/a up / n/a down
Network layer
LoRaWAN network-layer settings, behavior and session
Frequency plan *
Europe 863-870 MHz (SF9 for RX2 - recommended)
LoRaWAN version *
LoRaWAN Specification 1.0.1
Regional Parameters version *
TS001 Technical Specification 1.0.1
LoRaWAN class capabilities
Supports class B
Supports class C
Activation mode *
Over the air activation (OTAA)
Activation by personalization (ABP)
Define multicast group (ABP & Multicast)
Session and MAC state reset
Reset session and MAC state
Advanced MAC settings
Save changes

- Configure the "Uplink" and the "Downlink" for the "Payload" of the "End-device".

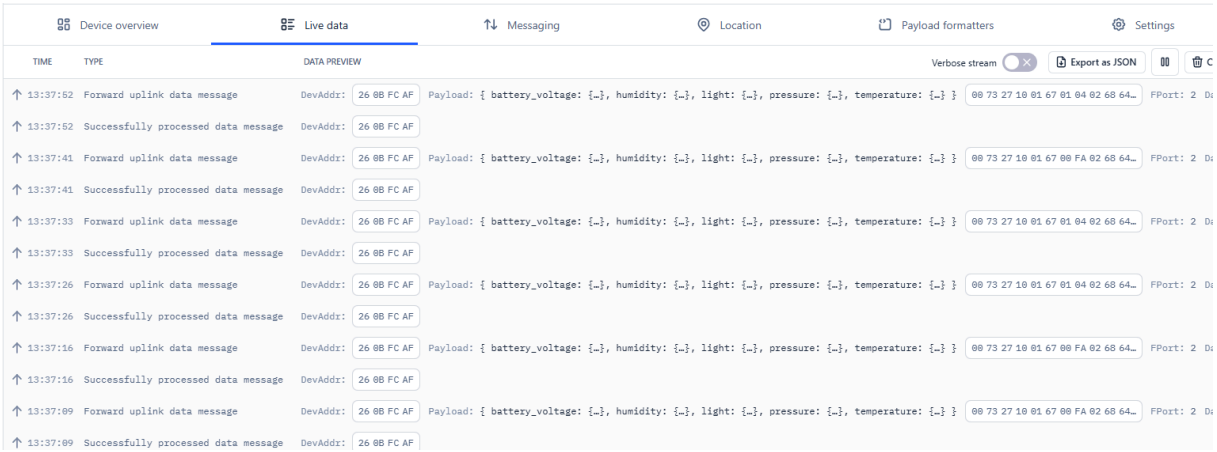




- Upload the program to the board

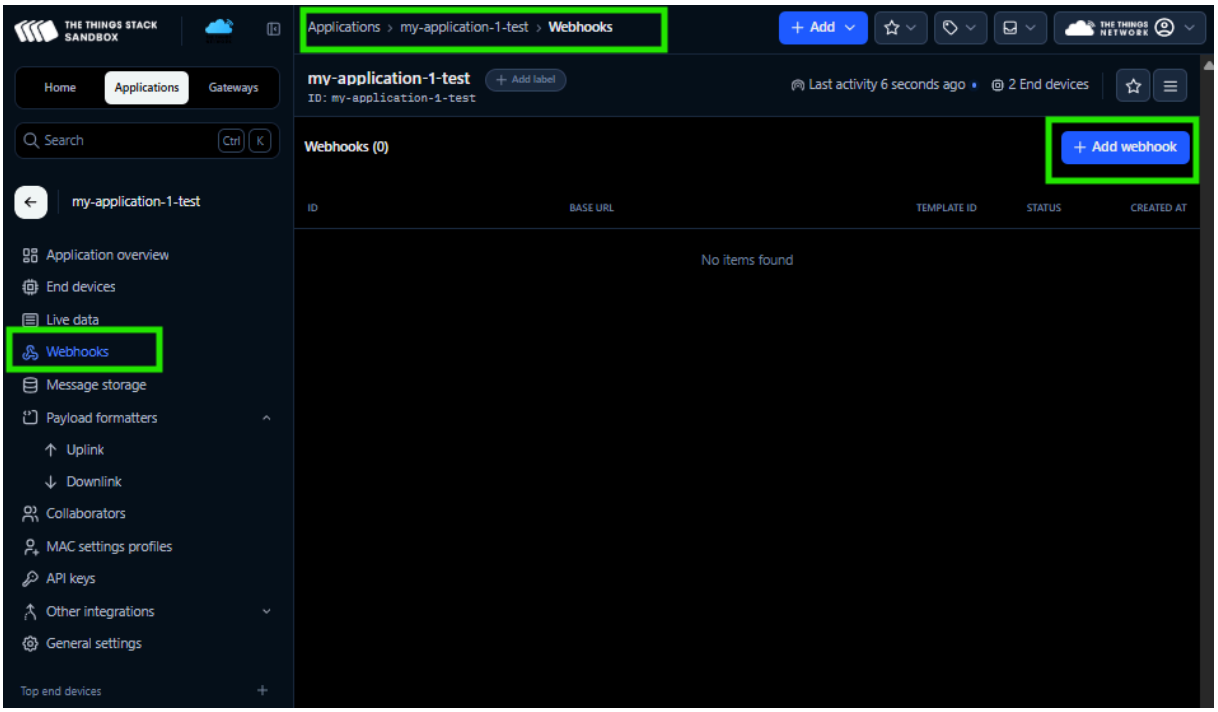


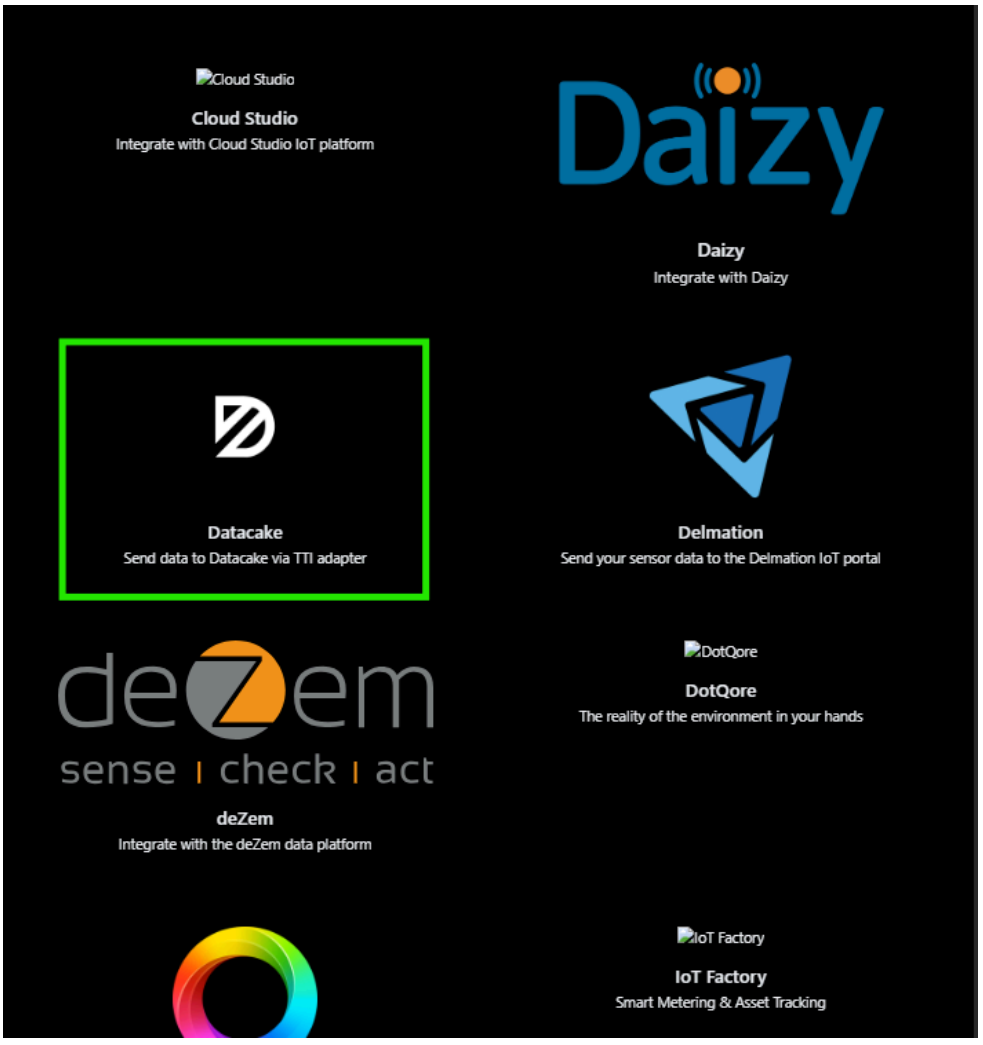
- Check if TTN is receiving the data; if so, turn off your development board.

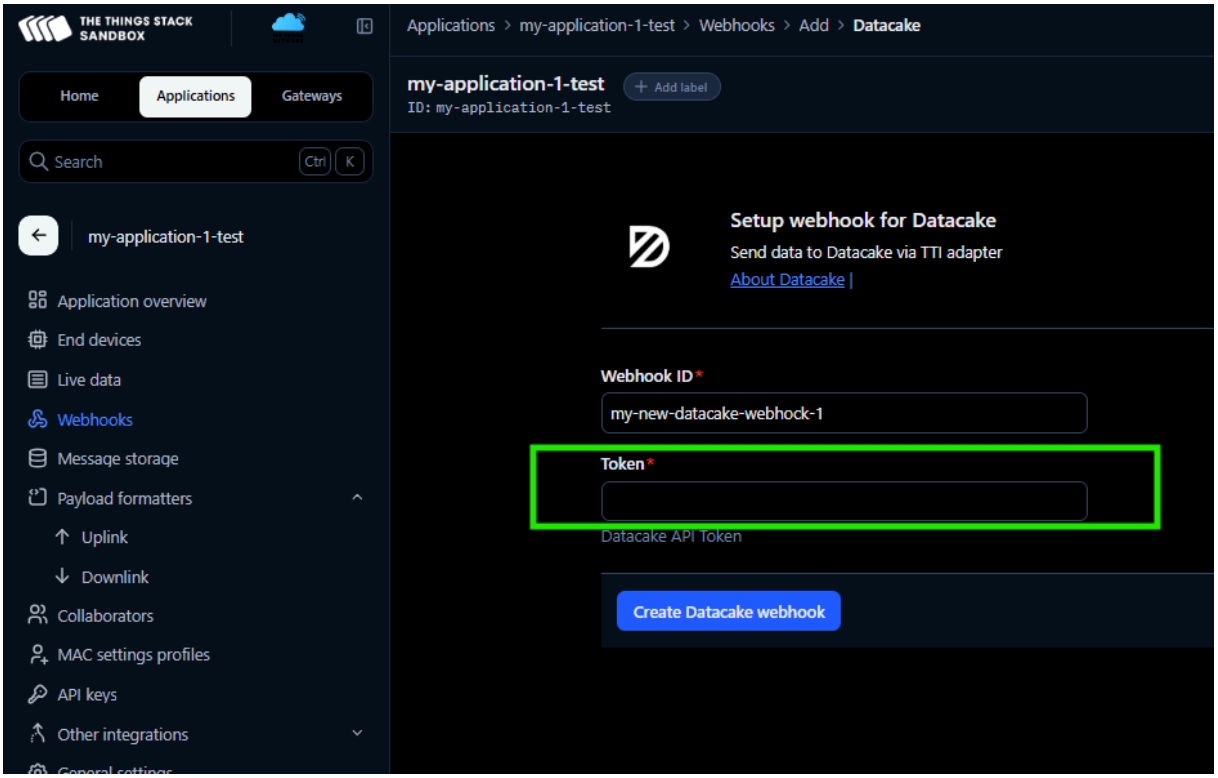


Uplink forwarding TTN to Datacake

- Setting up webhooks

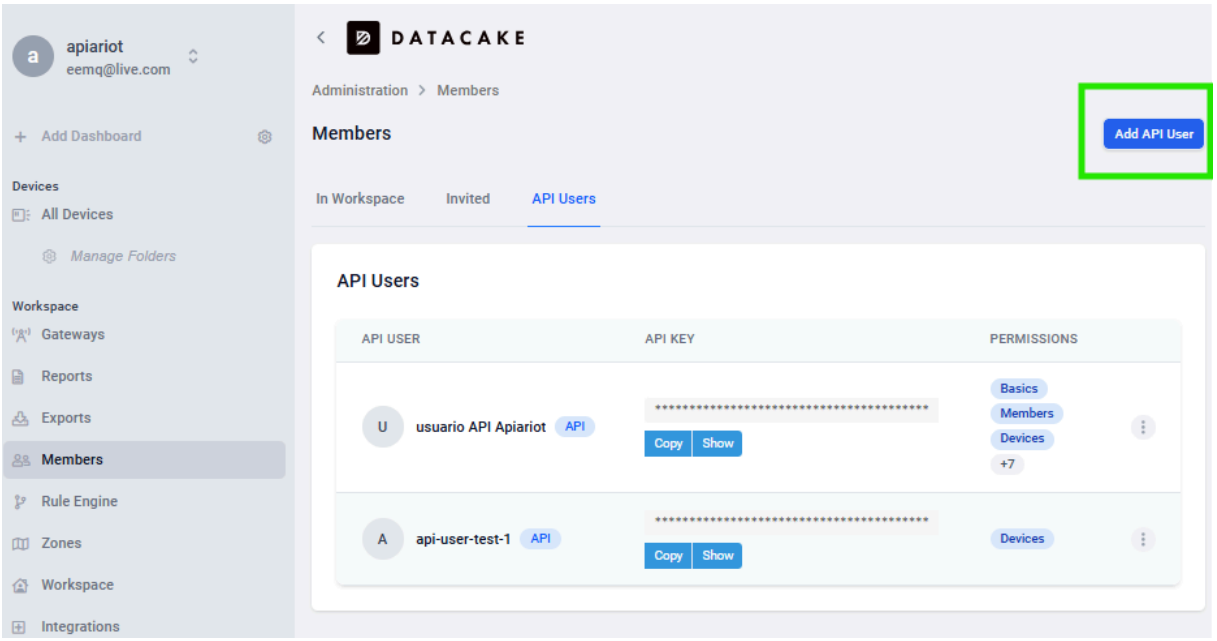


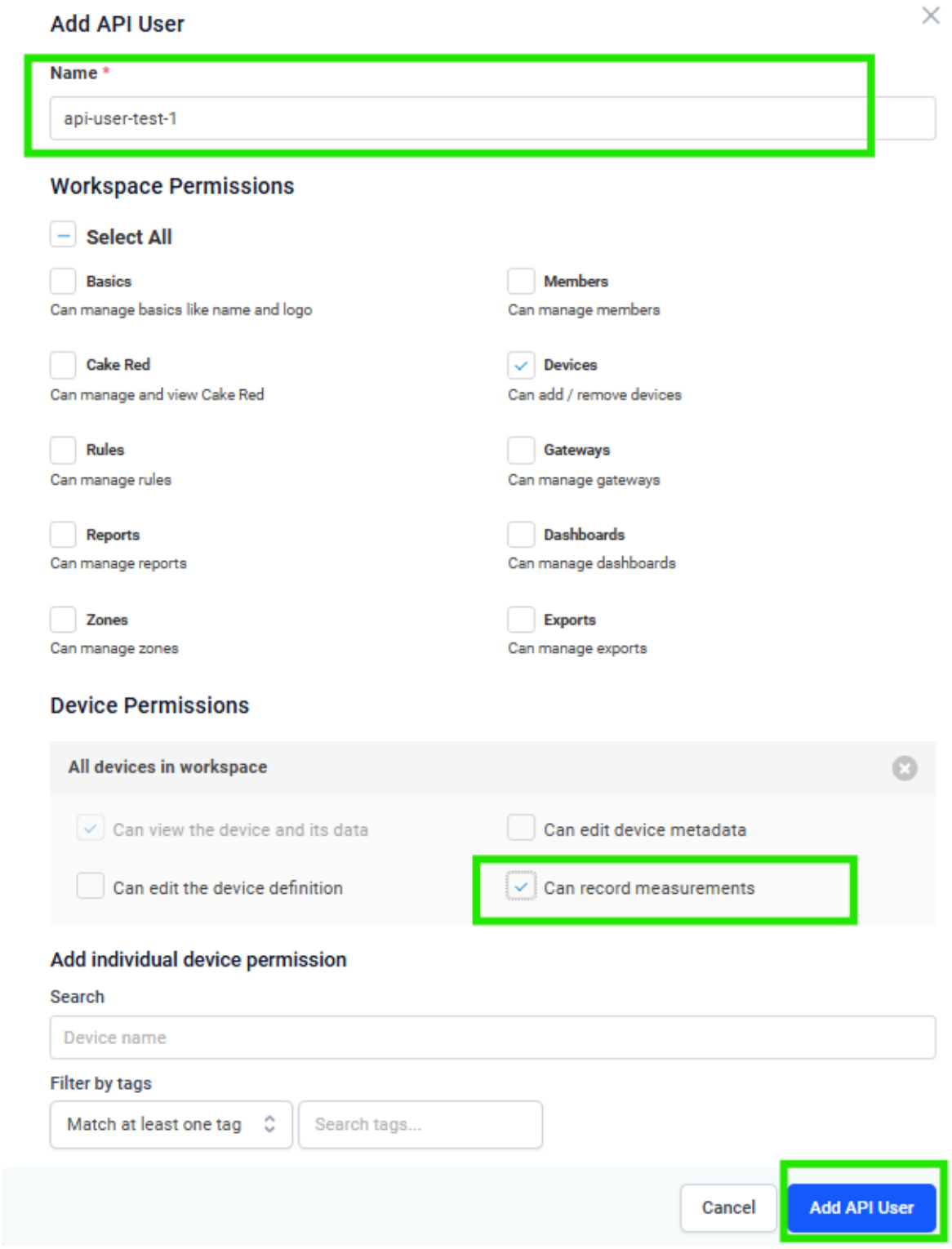




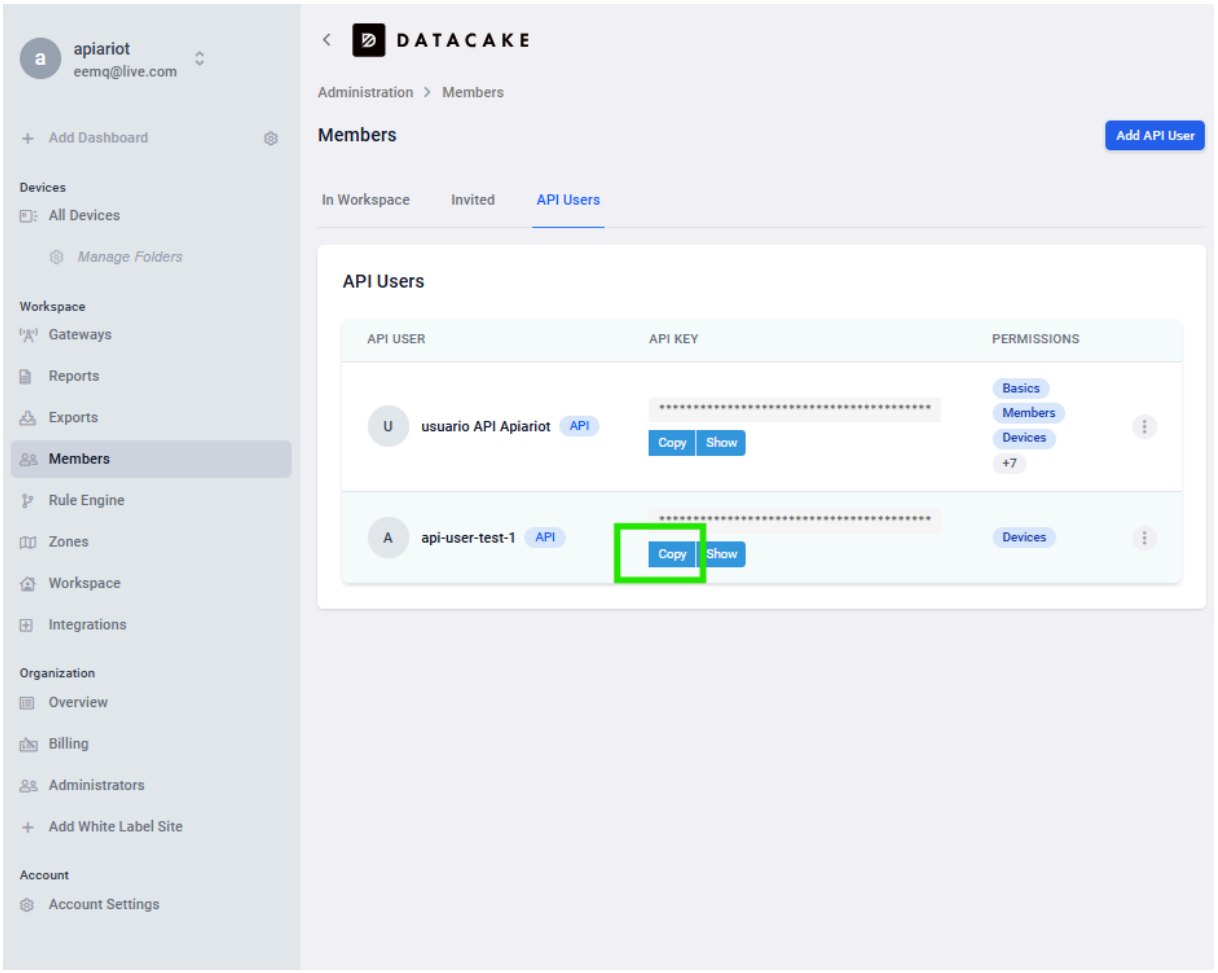
- You first need a Datacake account to get the token.
- Create your account in Datacake.

- And add an "API User"

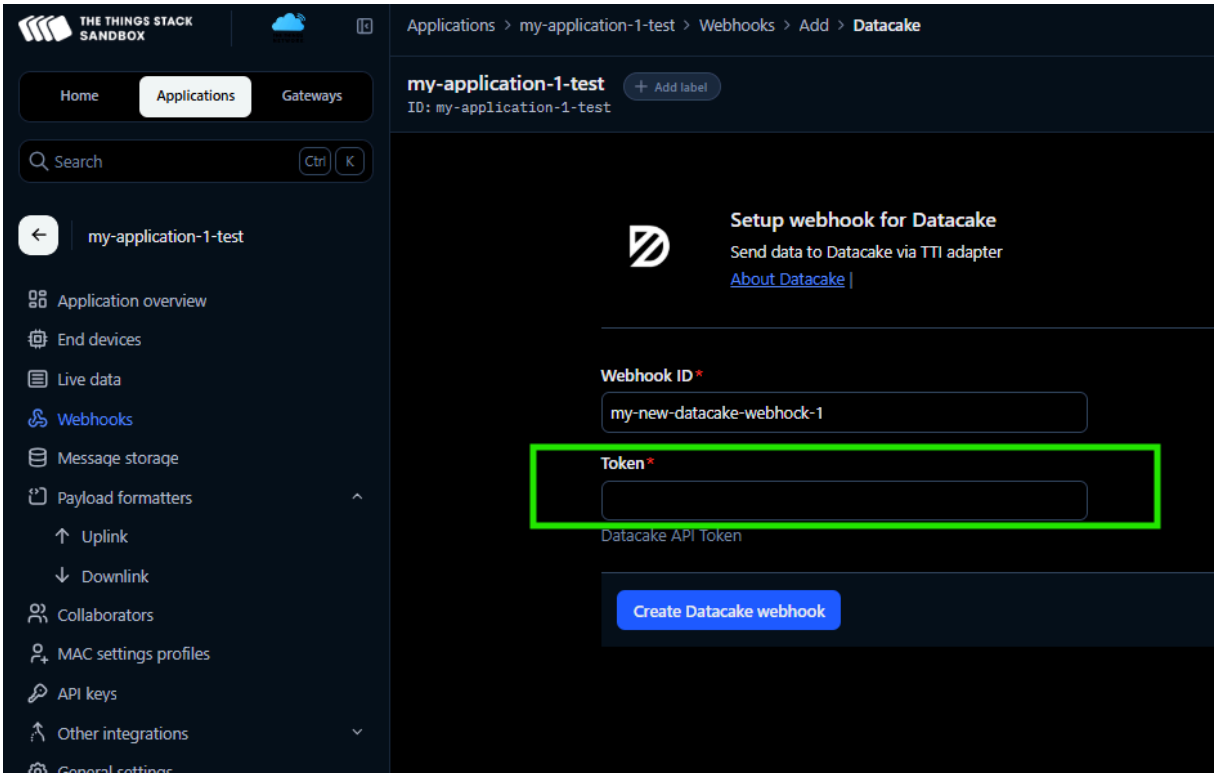



Add API User
Name *
api-user-test-1
Workspace Permissions
Select All
Basics
Can manage basics like name and logo
Members
Can manage members
Cake Red
Can manage and view Cake Red
Devices
Can add / remove devices
Rules
Can manage rules
Gateways
Can manage gateways
Reports
Can manage reports
Dashboards
Can manage dashboards
Zones
Can manage zones
Exports
Can manage exports
Device Permissions
All devices in workspace
Can view the device and its data
Can edit device metadata
Can edit the device definition
Can record measurements
Add individual device permission
Search
Device name
Filter by tags
Match at least one tag
Search tags...
Cancel
Add API User

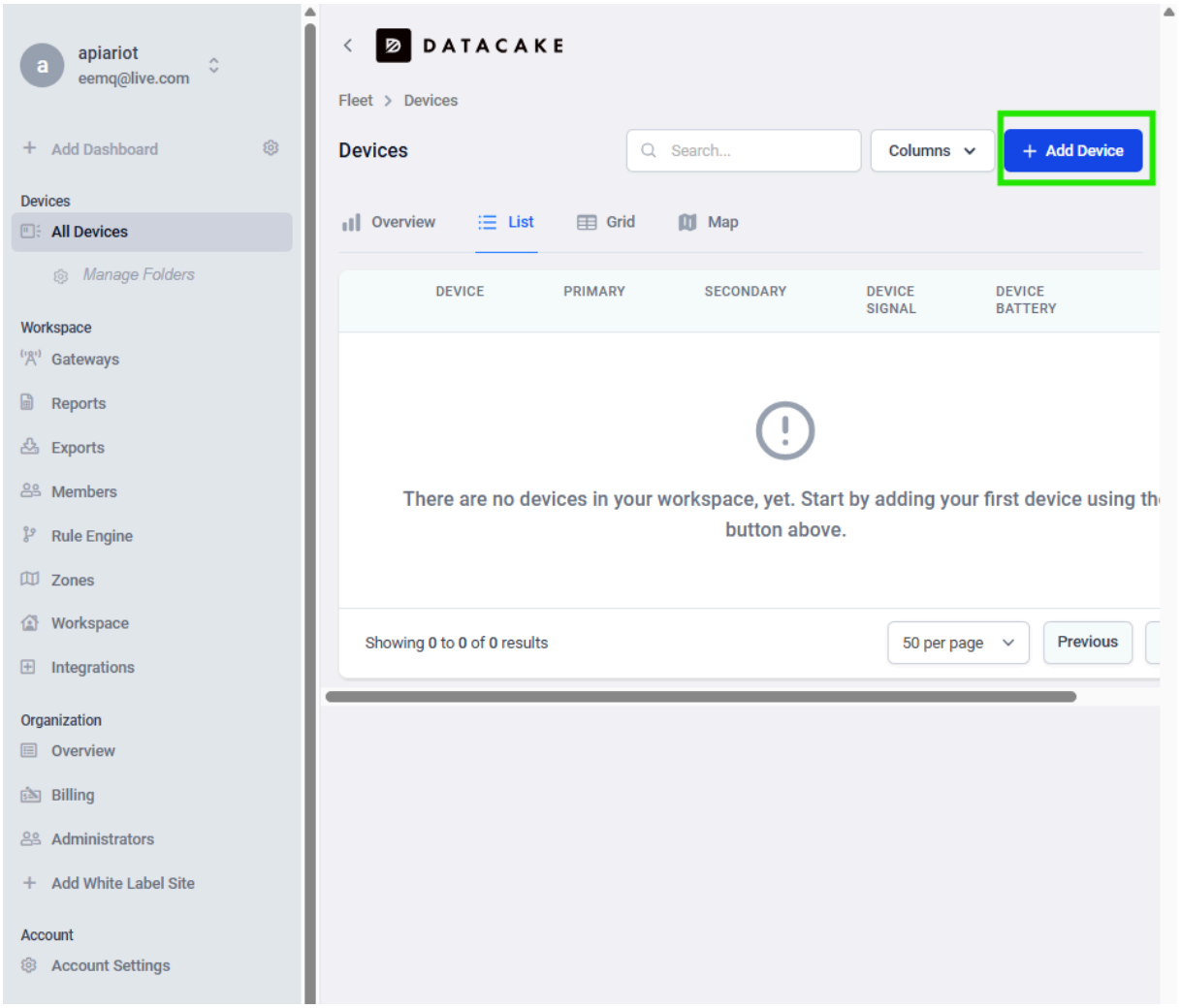
- Copy the token for TTN

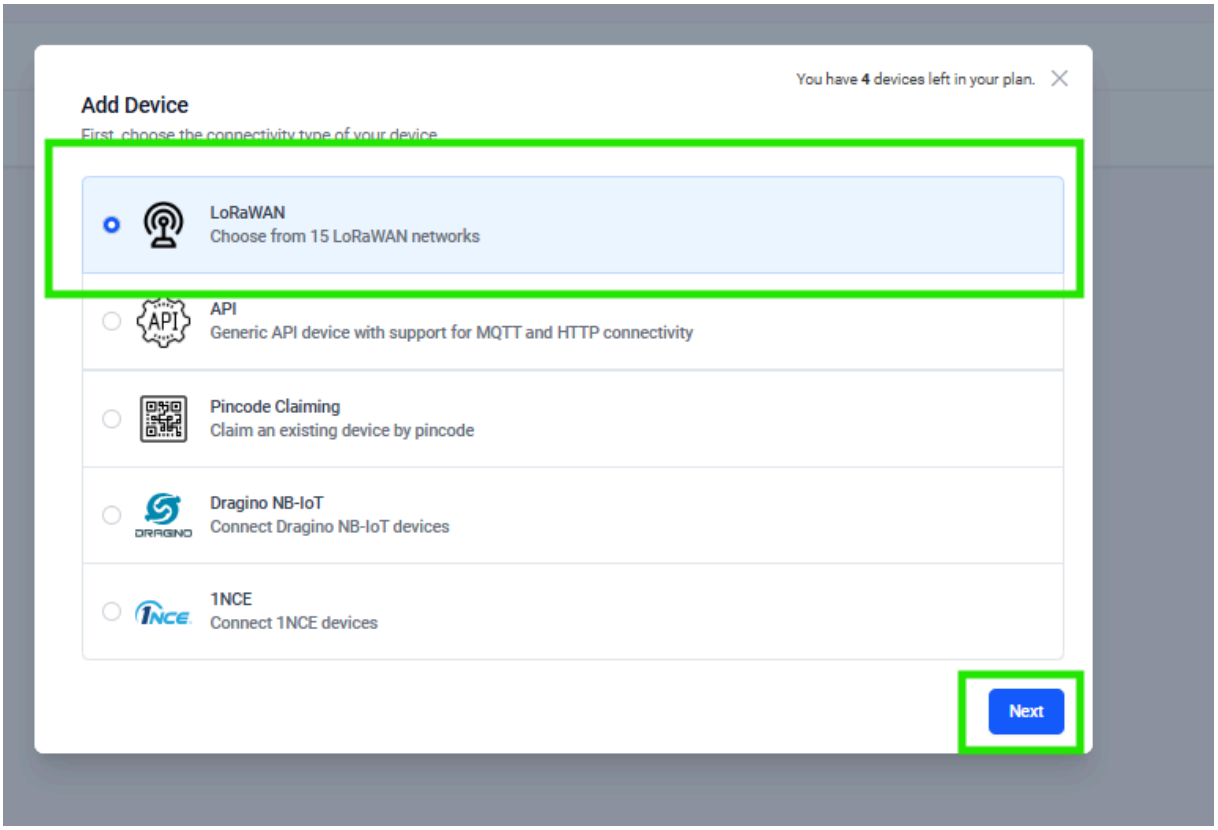


- You now have the token for TTN

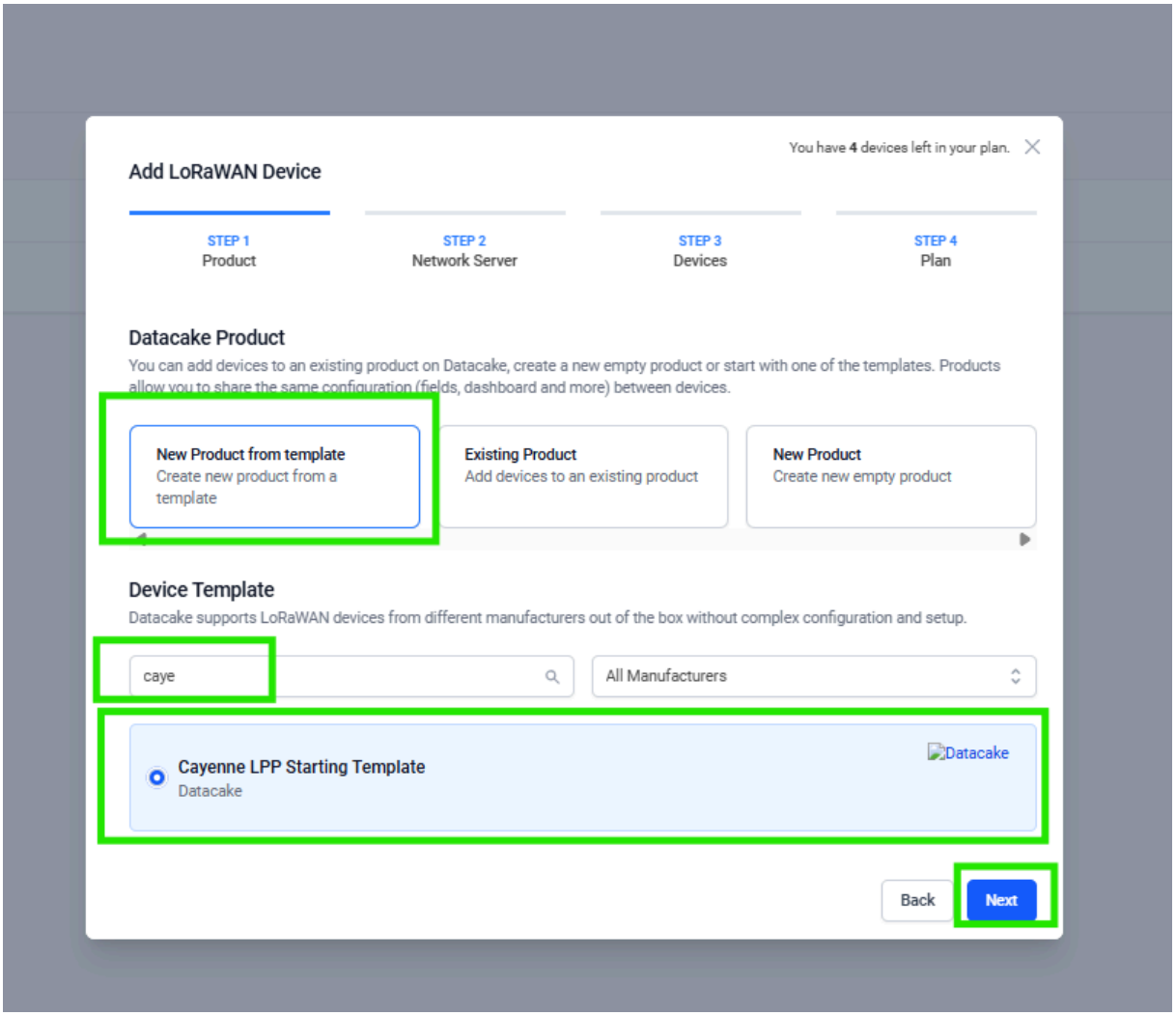


- Add your device now at Datacake




You have 4 devices left in your plan.
Add Device
First, choose the connectivity type of your device
LoRaWAN
Choose from 15 LoRaWAN networks
API
Generic API device with support for MQTT and HTTP connectivity
Pincode Claiming
Claim an existing device by pincode
Dragino NB-IoT
Connect Dragino NB-IoT devices
1NCE
Connect 1NCE devices
Next

- Use the network server: eu1.cloud.thethings.network (Switzerland)



SCHRITT 1
Produkt

SCHRITT 2
Netzwerk-Server

SCHRITT 3
Geräte

SCHRITT 4
Plan

Netzwerk-Server

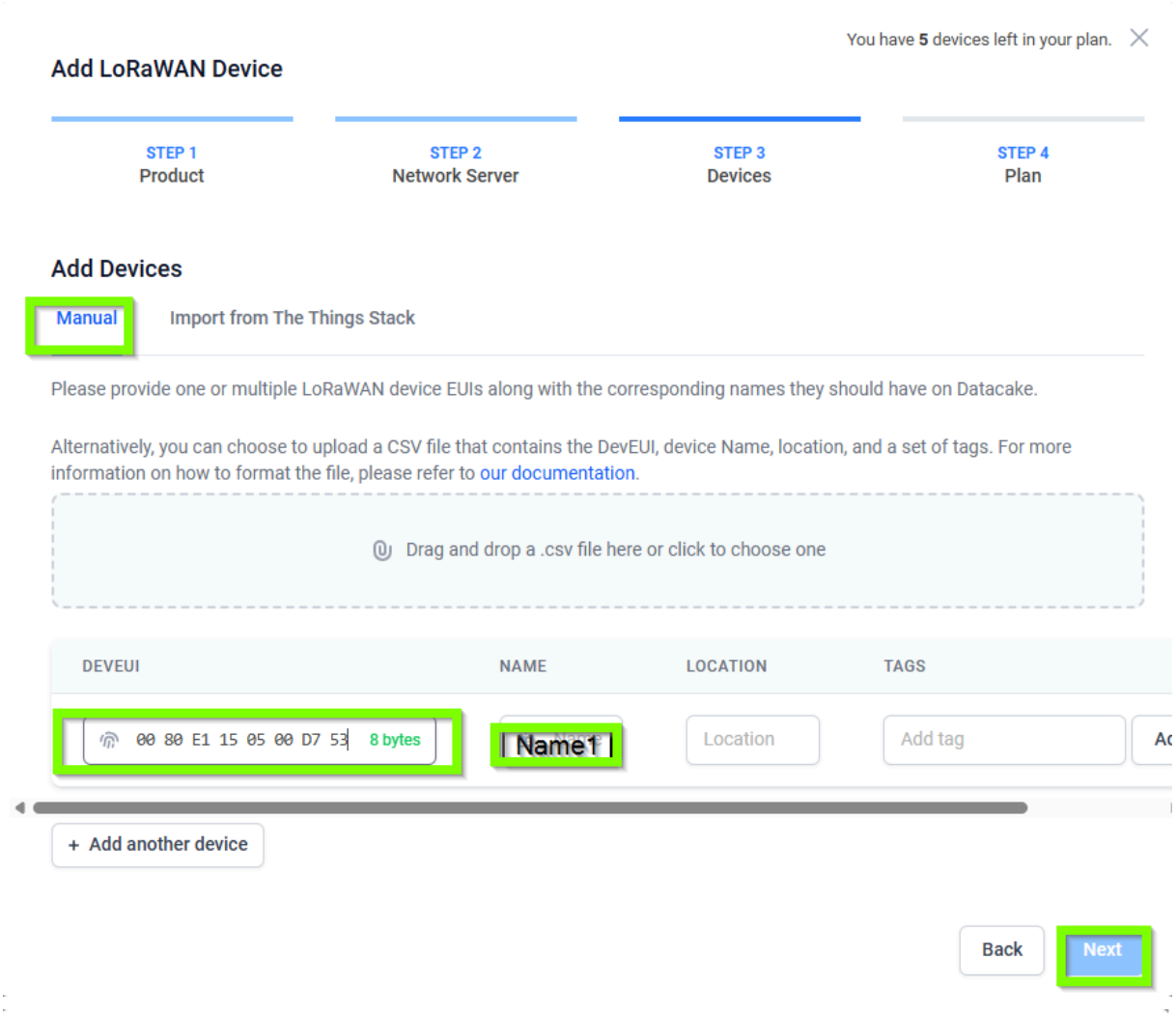
Bitte wählen Sie den LoRaWAN Network Server, mit dem Ihre Geräte verbunden sind.

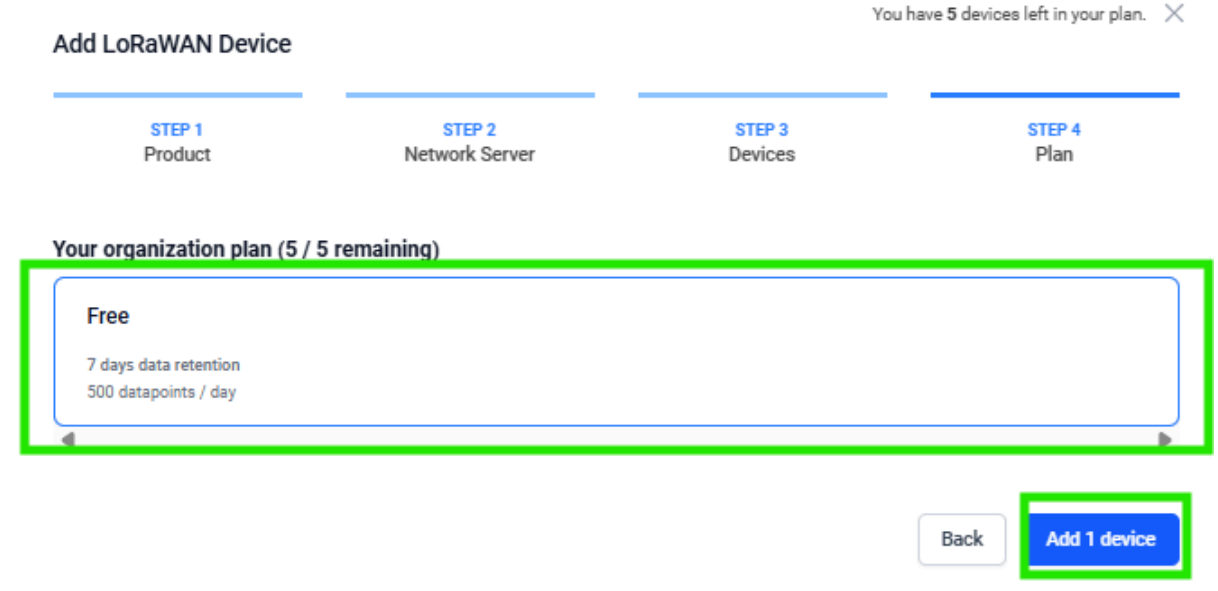
○	**Datacake LNS** AUTOMATIC SETUP Einfach starten und skalieren mit einem verwalteten LNS	Uplinks Downlinks
◉	**The Things Stack V3** TTN V3 / Things Industries	Uplinks Downlinks
○	**ThingPark**	Uplinks Downlinks
○	**KPN**	Uplinks Downlinks
○	**LORIOT**	Uplinks Downlinks
○	**ChirpStack**	Uplinks Downlinks
○	**Helium** Verwenden Sie Ihre eigene Konsole	Uplinks Downlinks

Anzeigen von **1** bis **6** von **14** Ergebnissen

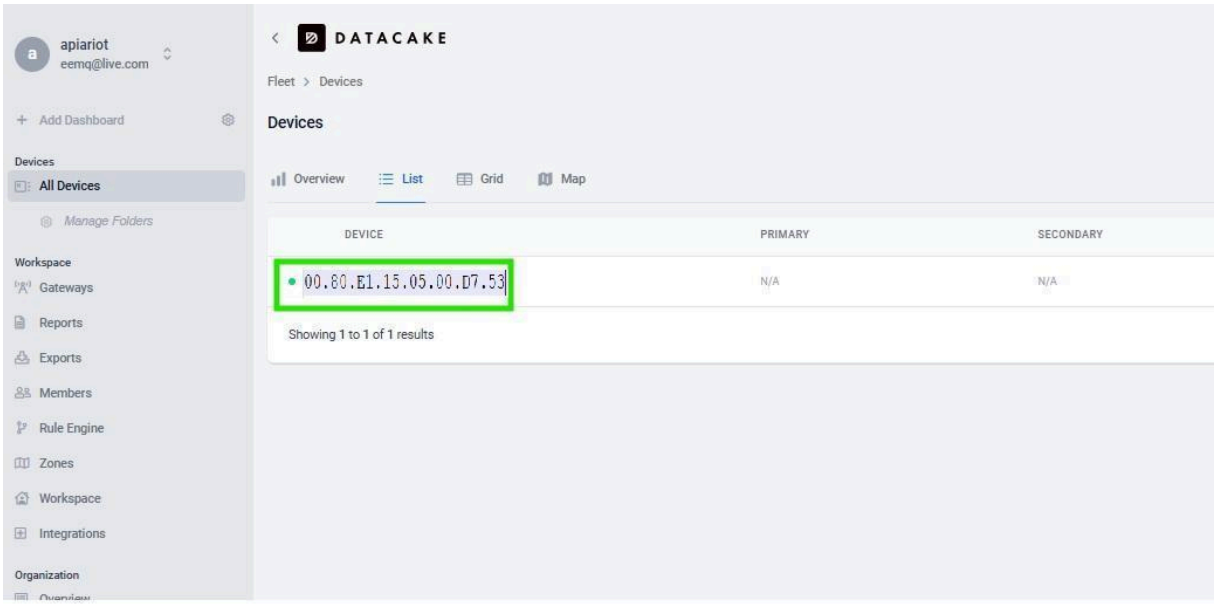
Vorherige Weiter

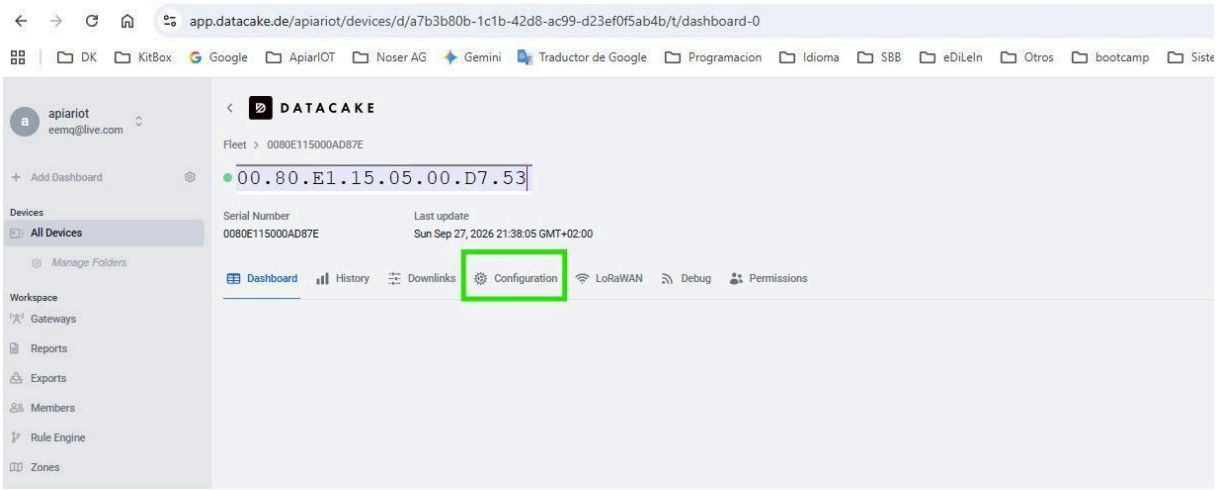
Zurück Weiter


You have 5 devices left in your plan.
Add LoRaWAN Device
STEP 1
Product
STEP 2
Network Server
STEP 3
Devices
STEP 4
Plan
Add Devices
Manual
Import from The Things Stack
Please provide one or multiple LoRaWAN device EUIs along with the corresponding names they should have on Datacake.
Alternatively, you can choose to upload a CSV file that contains the DevEUI, device Name, location, and a set of tags. For more information on how to format the file, please refer to our documentation.
Drag and drop a .csv file here or click to choose one
DEVEUI
NAME
LOCATION
TAGS
00 80 E1 15 05 00 D7 53
8 bytes
Name1
Location
Add tag
+ Add another device
Back
Next

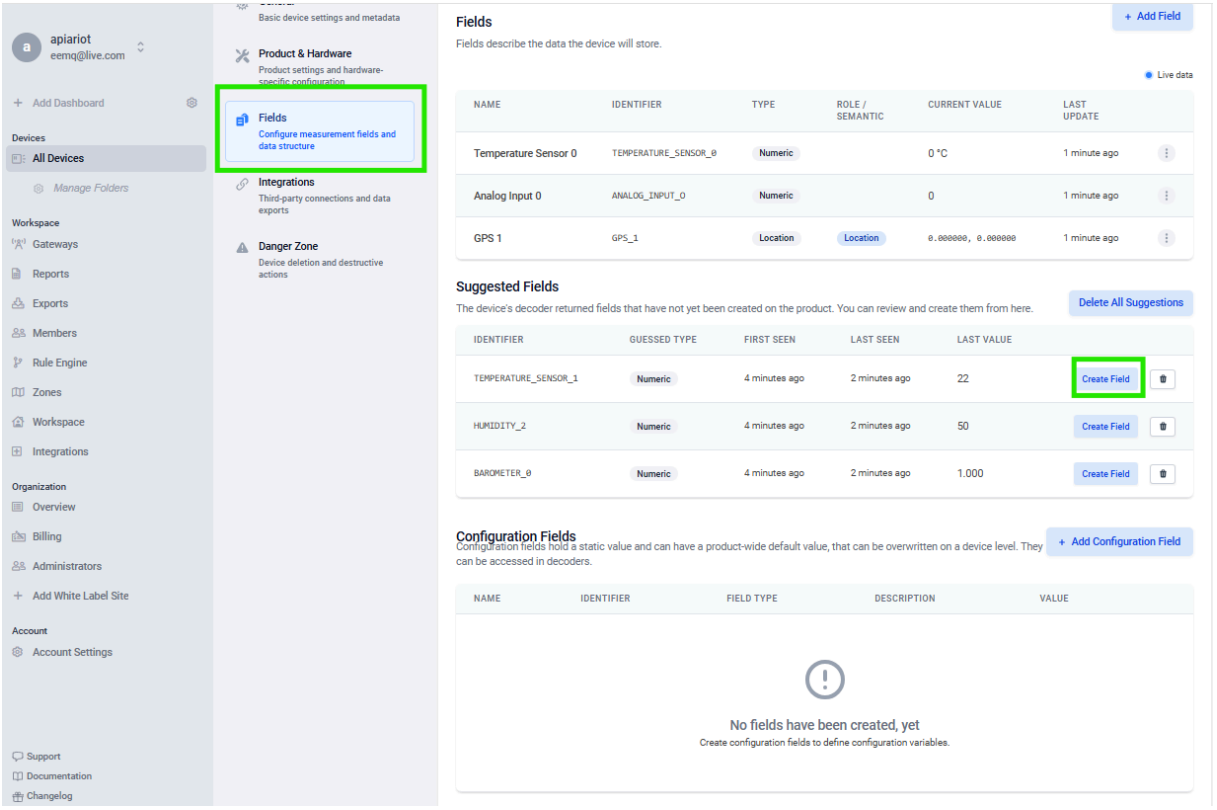

You have 5 devices left in your plan.
Add LoRaWAN Device
STEP 1
Product
STEP 2
Network Server
STEP 3
Devices
STEP 4
Plan
Your organization plan (5 / 5 remaining)
Free
7 days data retention
500 datapoints / day
Back
Add 1 device

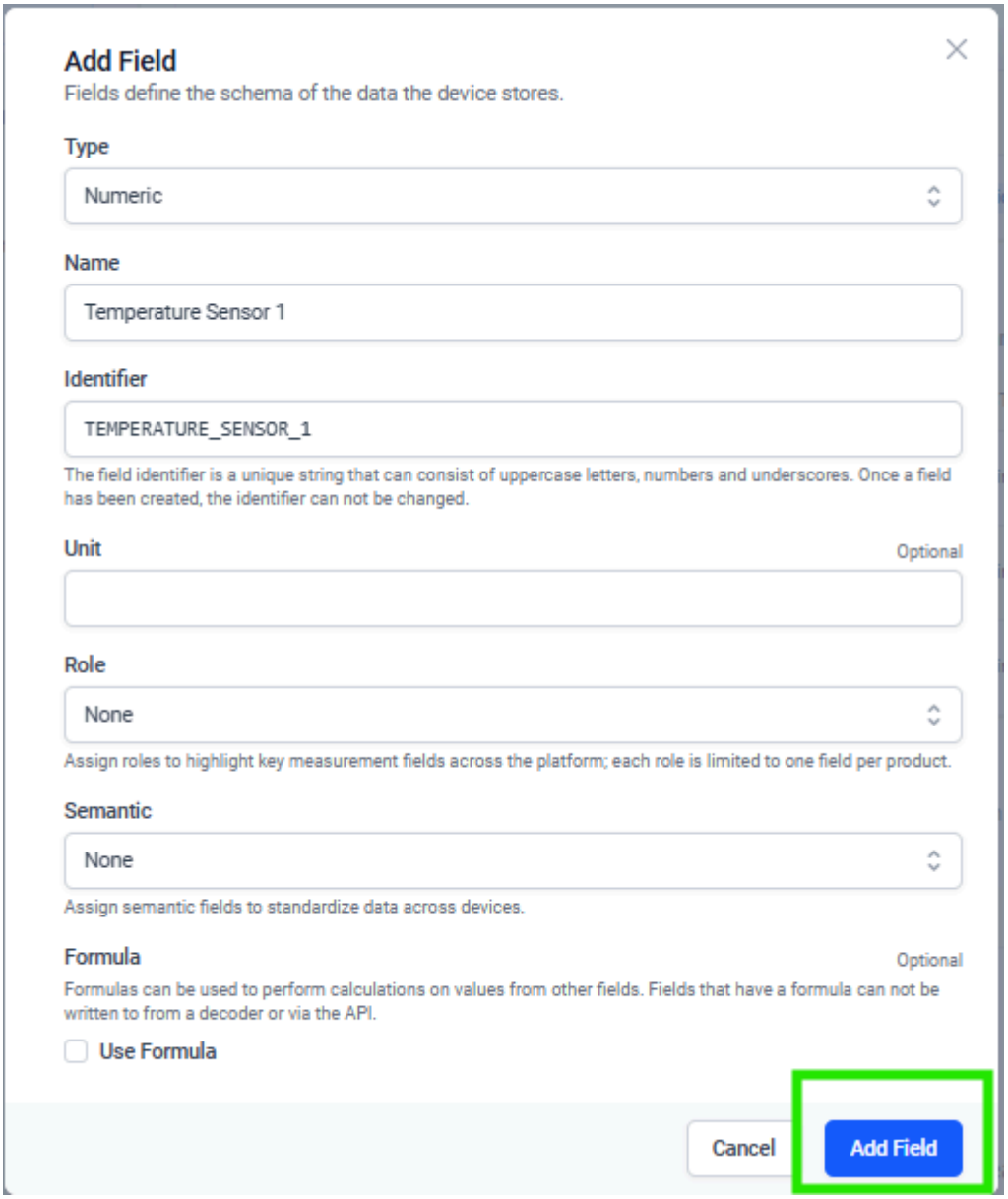
- Turn your "End Node" back on.



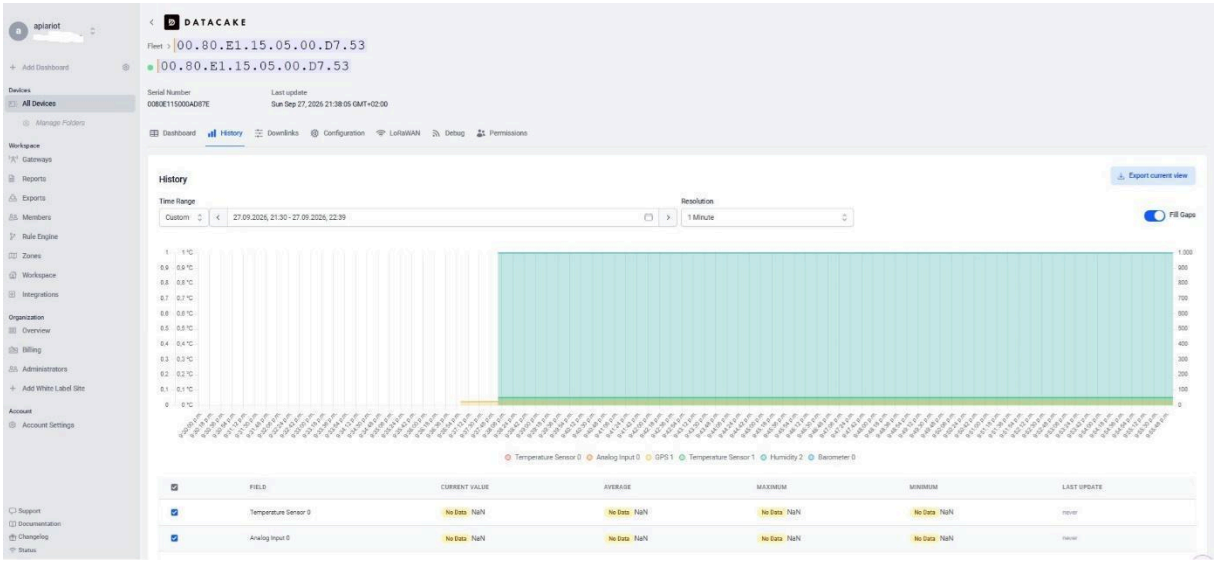


- Wait until data is received.





- You should now be able to see the data from your development board in Datacake.



- And in TTN as well

